

Chapter-04

Genetics and Genomics: The Molecular Basis of Heredity

BIO-3105; Biology for Engineers
Lecture-07+08; 29/03/2026
SM Rafiqul Islam, Ph.D.
Head, BGE, UIU.

Ref book: Biology for Engineers - Arthur T. Johnson
[2nd edition]

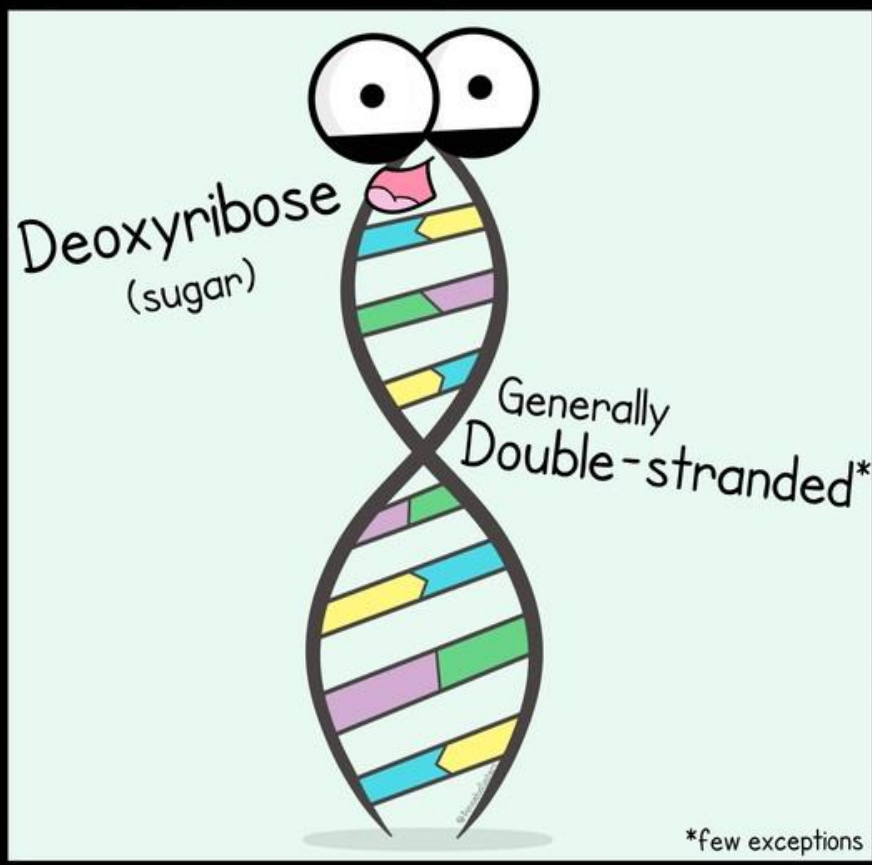
Biology for Engineers – G. K. Suraishkumar
and

https://www.youtube.com/watch?v=o_-6JXLYS-k



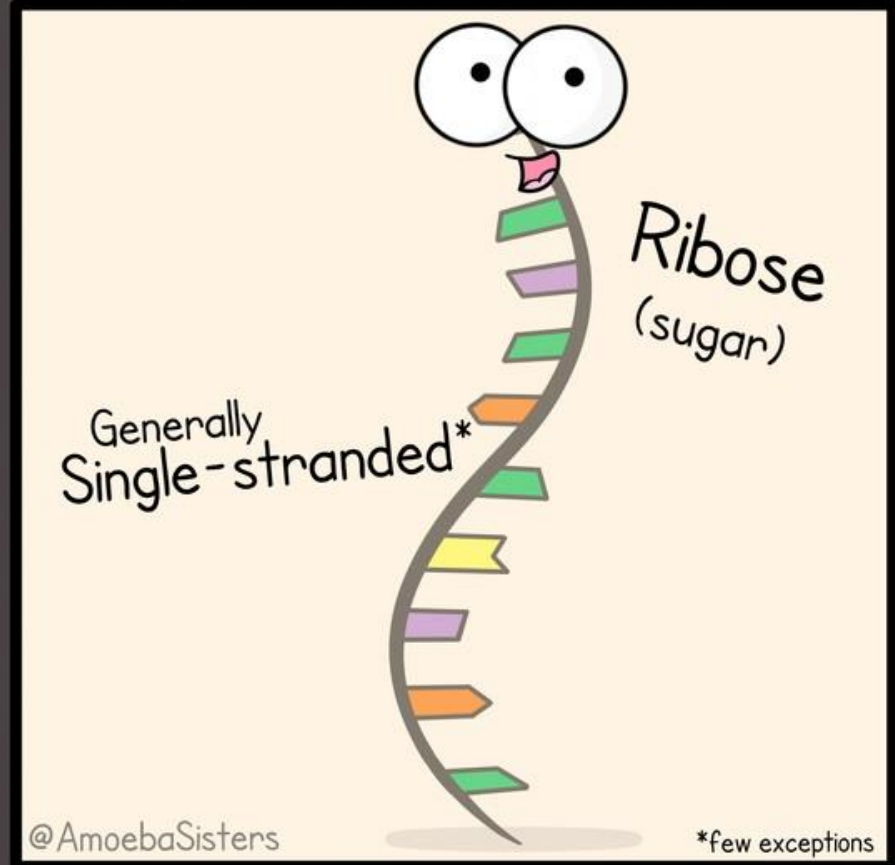
Nucleic Acids

DNA (Deoxyribonucleic Acid)



- Adenine
- Thymine
- Cytosine
- Guanine

RNA (Ribonucleic Acid)



- Adenine
- Uracil
- Cytosine
- Guanine

The Discovery of DNA

Erwin Chargaff (1951):

- Rule of Base pairing



Rosalind Franklin & Maurice Wilkins (1953):

- X-ray diffraction pattern of DNA



James Watson & Francis Crick (1953):

- Molecular structure of DNA



James Watson



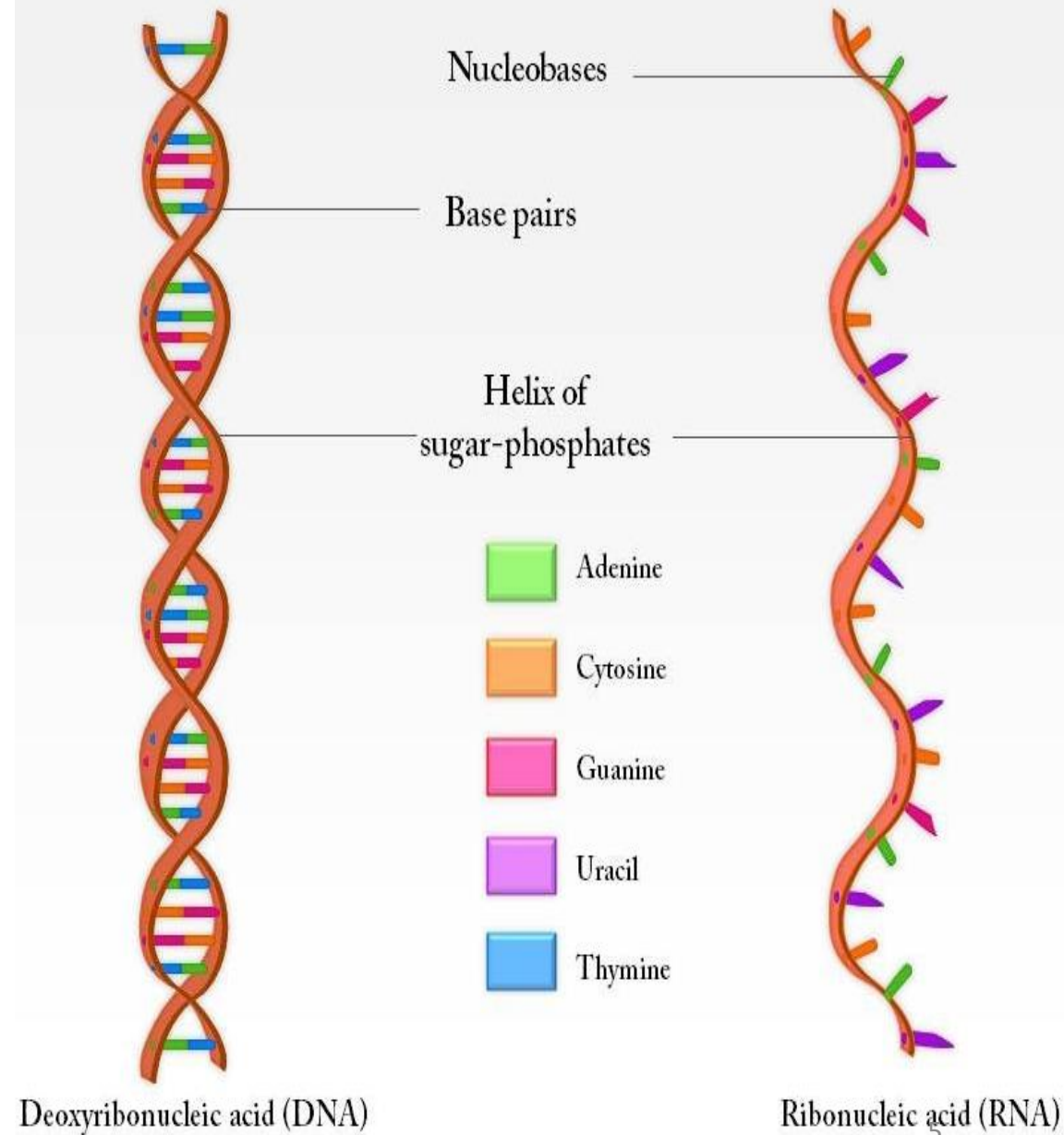
Francis Crick

Basics about Nucleic Acids

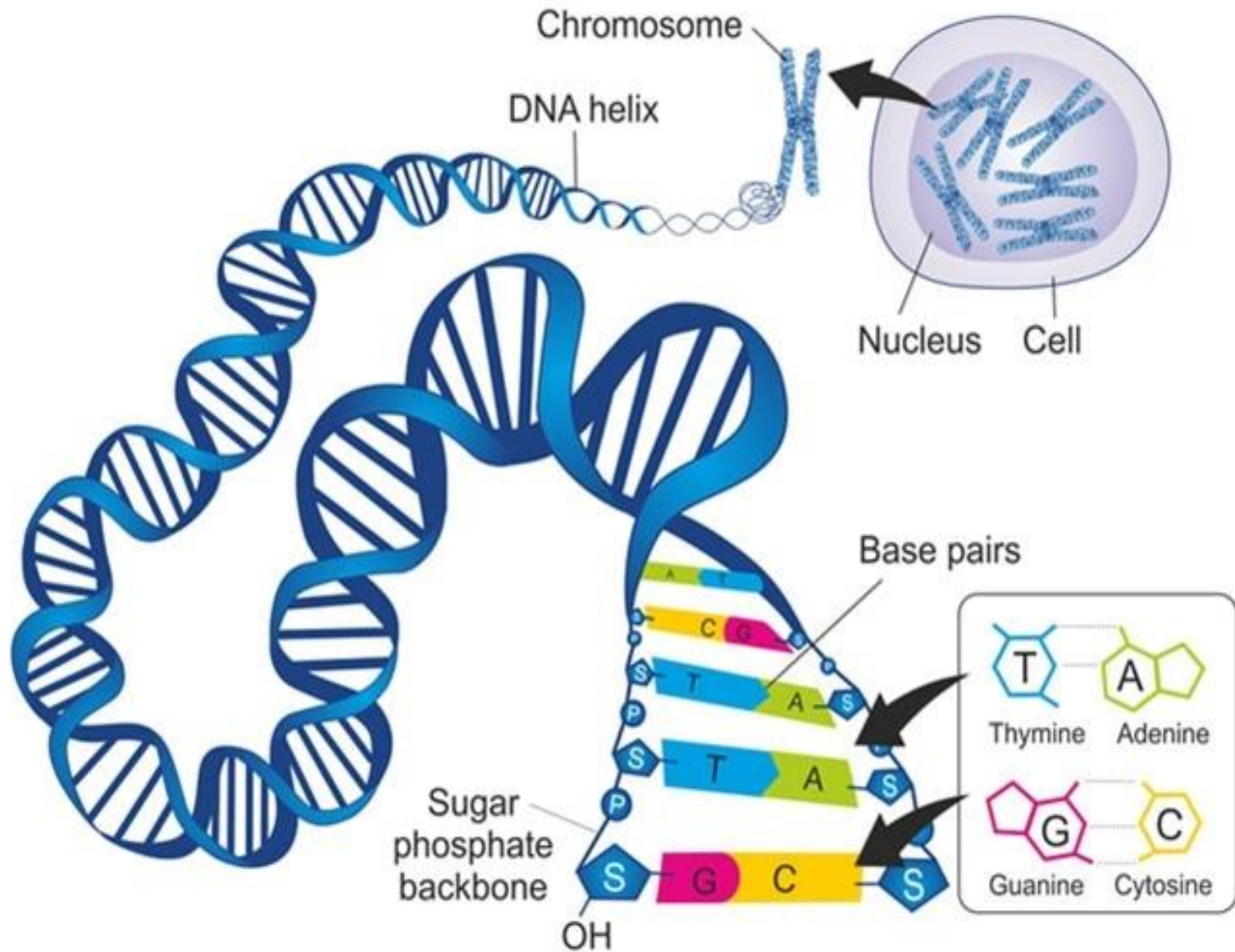
- Nucleic acids carry the genetic blueprint of a cell, and carry instructions for functioning of the cell
- Two types: DNA (deoxyribonucleic acid) and RNA (ribonucleic acid)
- DNA makes up the **genetic content (genome)** of cells
- Different types of RNA; messenger RNA (**mRNA**), transfer RNA(**tRNA**), ribosomal RNA(**rRNA**).
- Nucleic acids are made up of monomers called nucleotides

DNA and RNA

- Genetics mainly deals with the study of genes, heredity, and genetic variation.
- Genes exist in chromosomes and chromosomes are comprised of **DNA** and proteins. DNA is a molecule that carries genetic information in all living organisms where it is used in function, growth, development and reproduction.
- Deoxyribonucleic acid (DNA) and ribonucleic acid (RNA) are the two types of nucleic acids found in living systems.
- DNA acts as the genetic material in most of the organisms. RNA though it also acts as a genetic material in some viruses, mostly functions as a messenger.



DNA



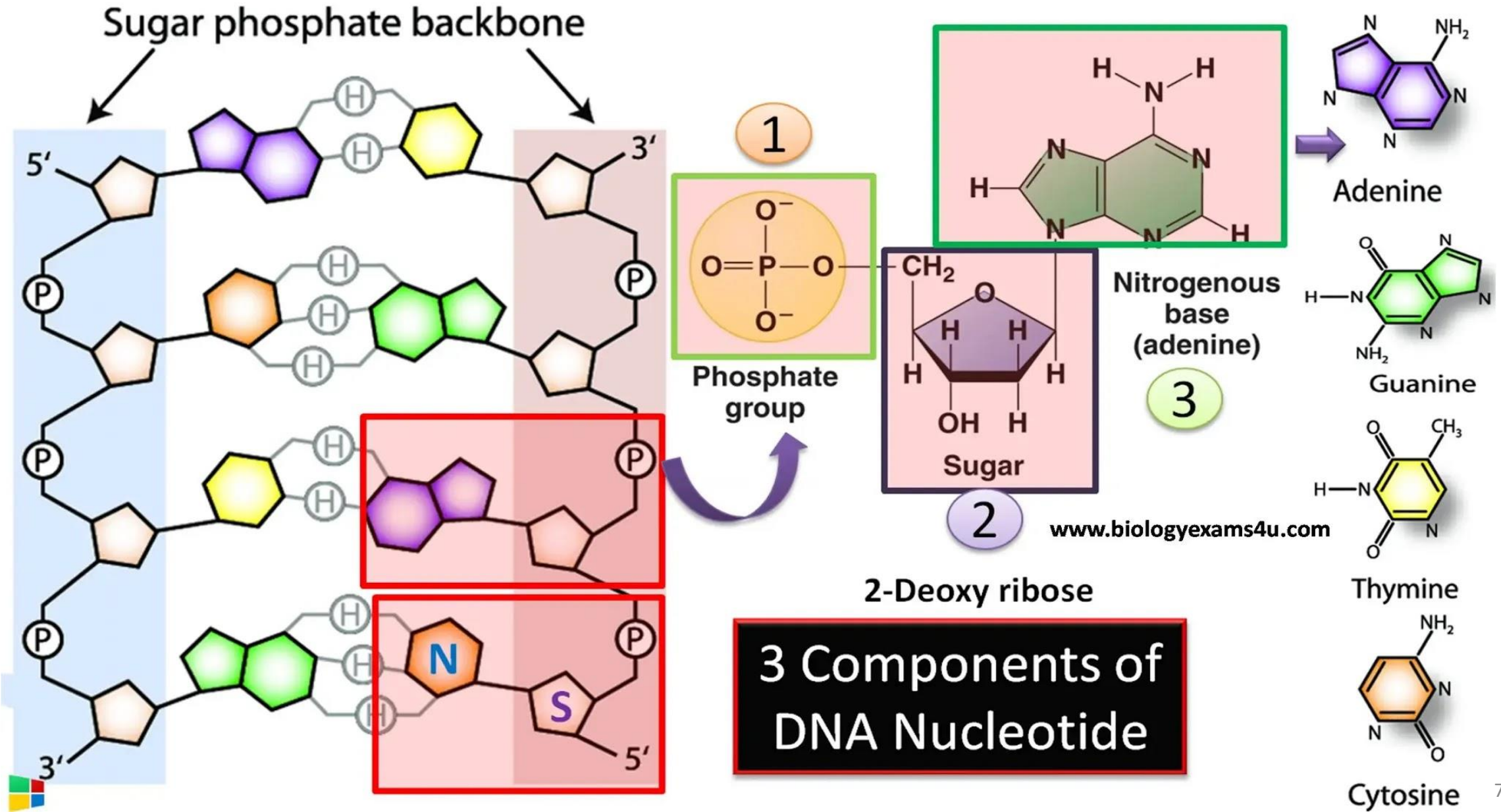
DNA is a double-helical structure that carries all the genetic information.

Its **length is determined by the number of nucleotide pairs present in it.**

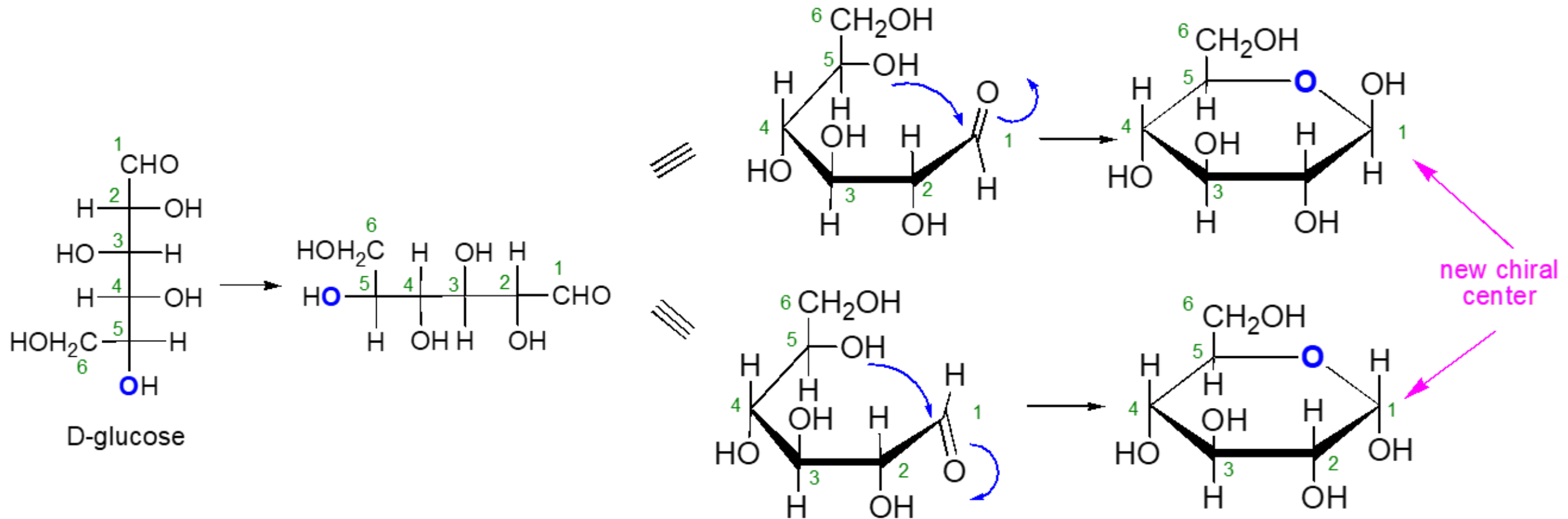
It is an **acidic substance** in the nucleus identified by Friedrich Meischer.

Its double helical structure was given by **Watson and Crick.**

DNA

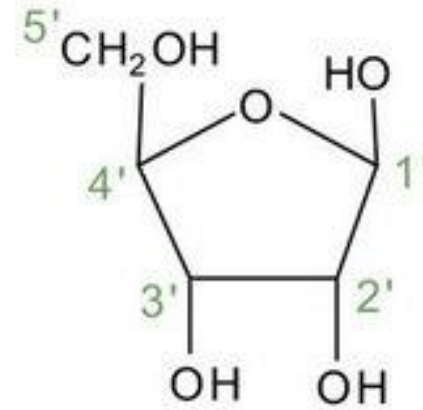
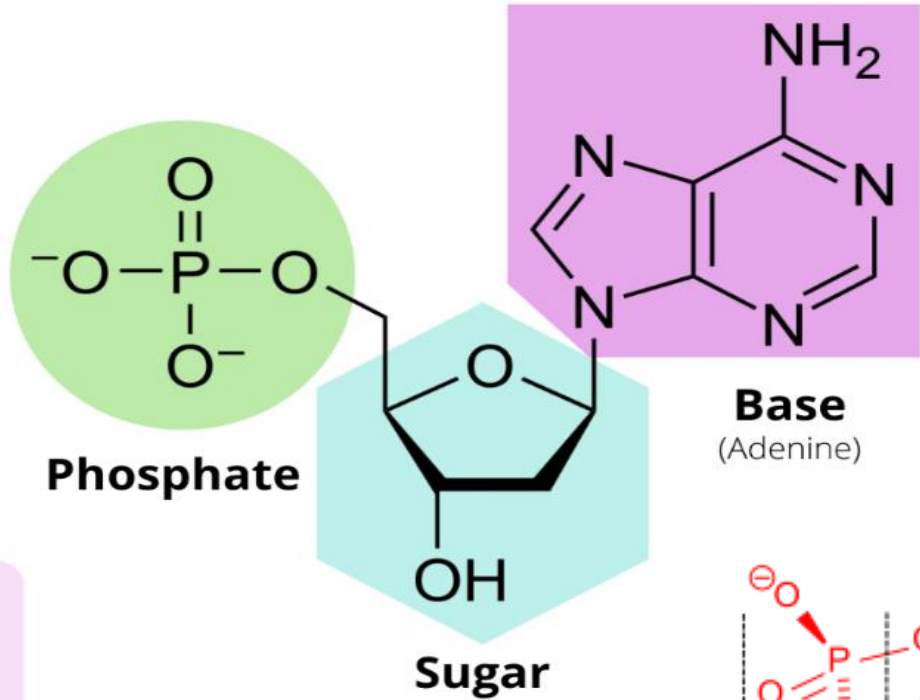


Cyclic Forms of Carbohydrates: Base Sugar

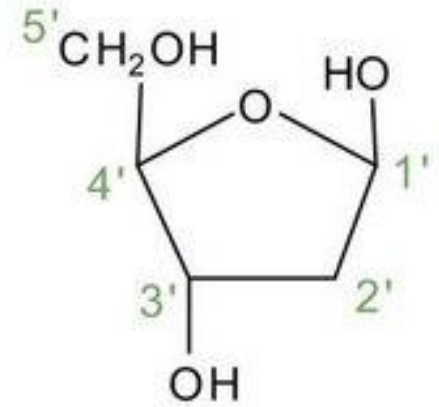


3 parts of DNA and RNA

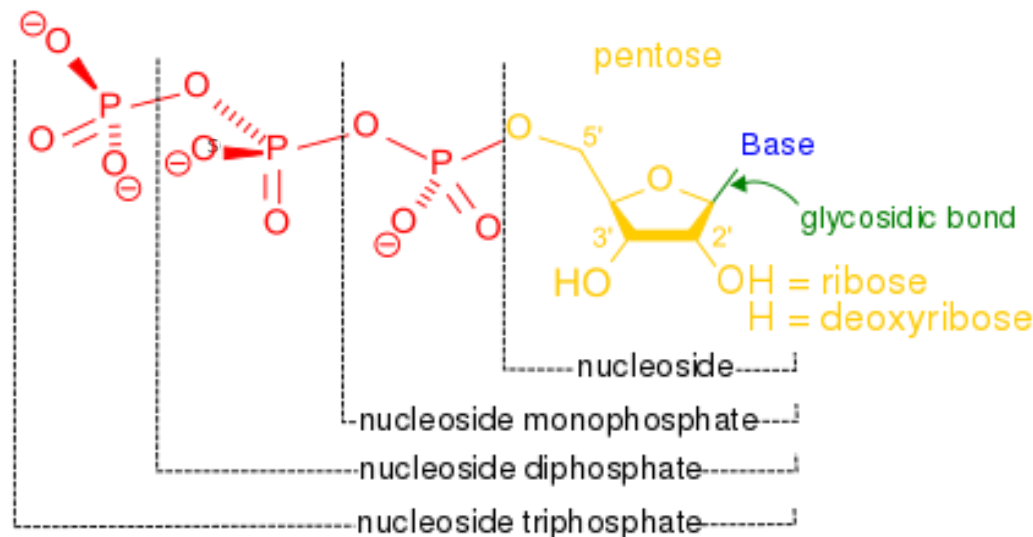
3 Parts of a Nucleotide



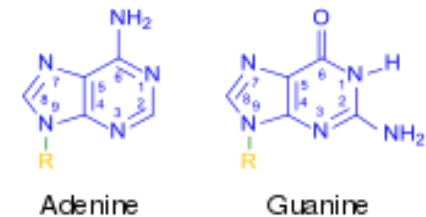
β -Ribose
(found in RNA)



β -2-Deoxyribose
(found in DNA)



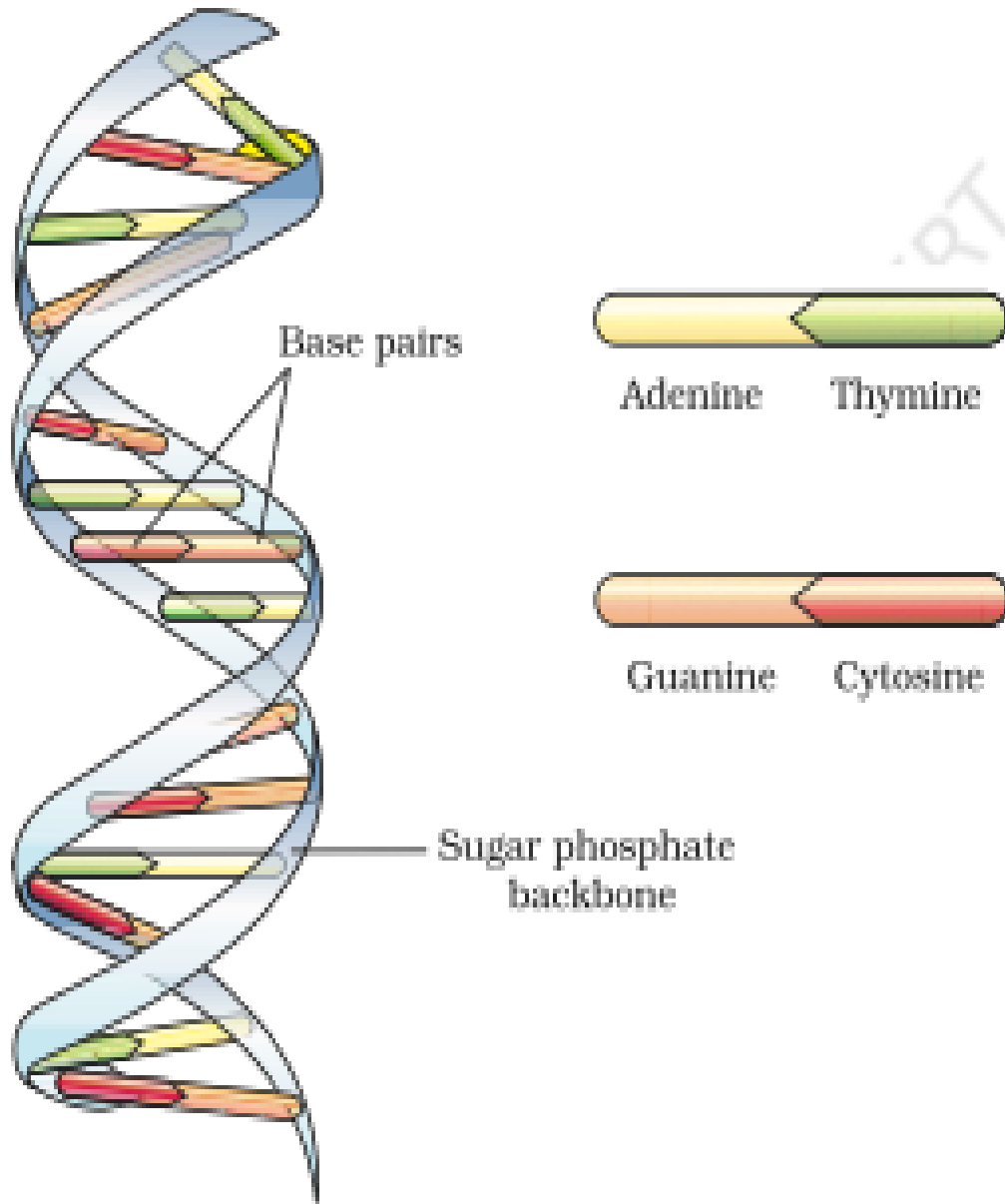
Purines



Pyrimidines



DNA



- DNA is made up of 6 molecular structures that comprise of one phosphate molecule and five carbon sugar termed deoxyribose.
- A nucleotide is a basic building block of DNA. A nucleotide is comprised of one of the 4 bases, one sugar molecule, and one phosphate molecule. A sugar-phosphate chain act as a backbone and bases are on the inside. Nucleotide subunits are linked together to form a DNA strand thus providing polar stability.
- The three-dimensional structure of DNA arises from chemical and structural features of 2 polynucleotide chain. A purine base pairs up with pyrimidine base. For instance guanine pairs with cytosine. So the two strands that are held together by a hydrogen bond are complementary to each other and they run in the antiparallel direction.

DNA

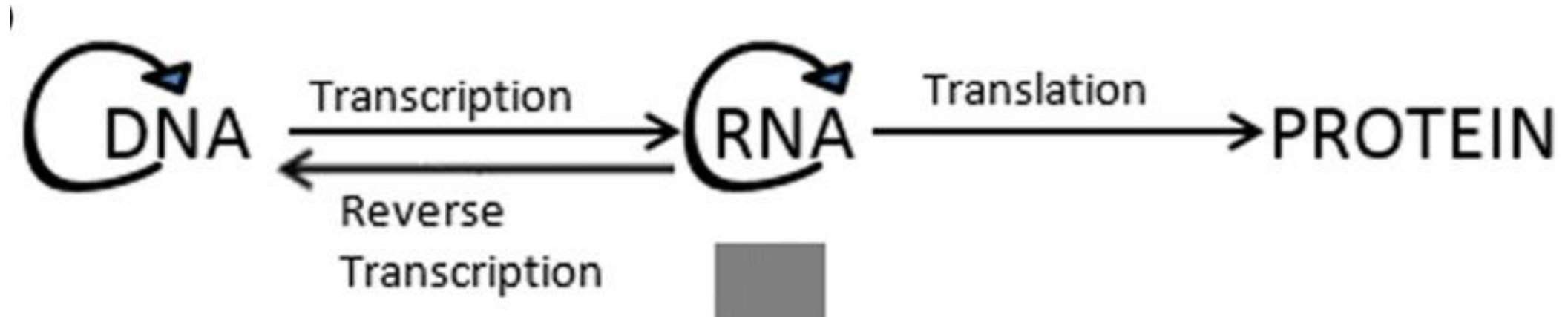
The salient features of the Double-helix structure of DNA are as follows:

- It is made of two polynucleotide chains, where the backbone is constituted by sugar-phosphate, and the bases project inside.
- The two chains have anti-parallel polarity. It means, if one chain has the polarity *5' to 3'*, the other has *3' to 5'*.
- The bases in two strands are paired through hydrogen bond (H-bonds) forming base pairs (bp). Adenine forms two hydrogen bonds with Thymine from opposite strand and vice-versa. Guanine is bonded with Cytosine with three H-bonds. As a result, always a purine comes opposite to a pyrimidine. This generates approximately uniform distance between the two strands of the helix.
- The two chains are coiled in a right-handed fashion. The pitch of the helix is 3.4 nm and there are roughly 10 bp in each turn. Consequently, the distance between a bp in a helix is approximately 0.34 nm.
- The plane of one base pair stacks over the other in double helix. This, in addition to H-bonds, confers stability of the helical structure.

Central Dogma

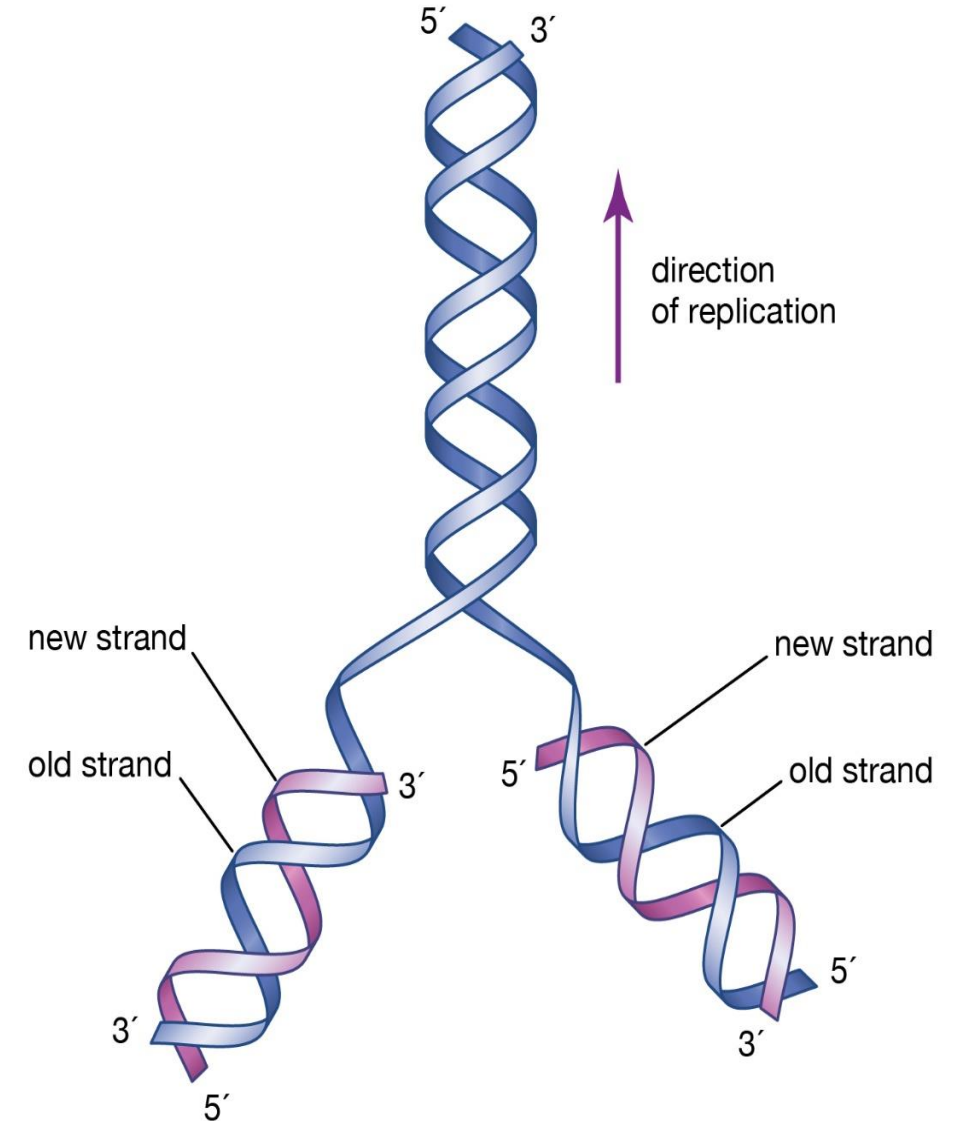
The central dogma was proposed by Crick. The central dogma states that the DNA is converted into RNA and the RNA is converted into proteins.

In retroviruses, the **flow of information is opposite**, i.e., RNA to DNA to mRNA to Protein.



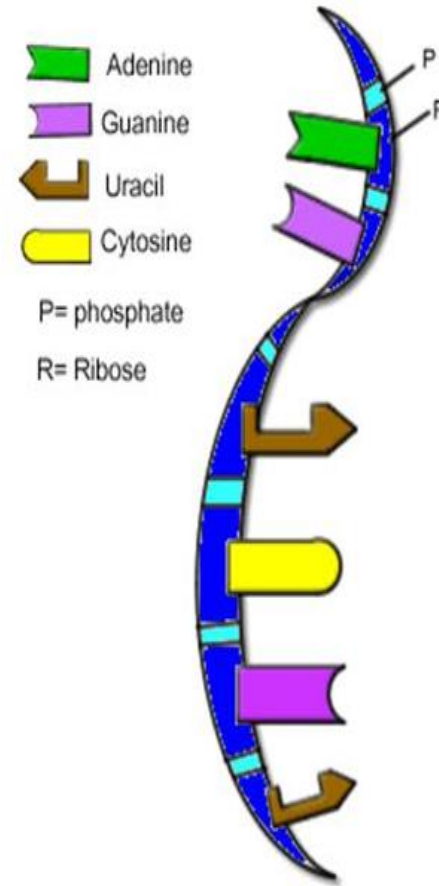
DNA Replication

- DNA is self-replicative.
- It occurs in the S-phase of the life cycle.
- It takes only a few minutes in prokaryotes but hours in eukaryotes.
- DNA undergoes semi-conservative replication, i.e., two strands of DNA are formed. One strand is the same as one of the strands while the other is complementary to the parent strand.
- The replication occurs in 5'-3' direction.



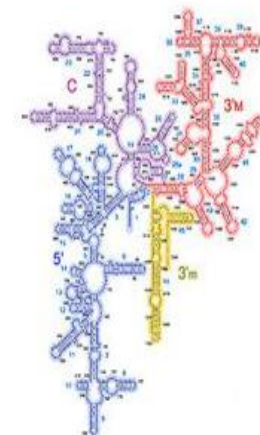
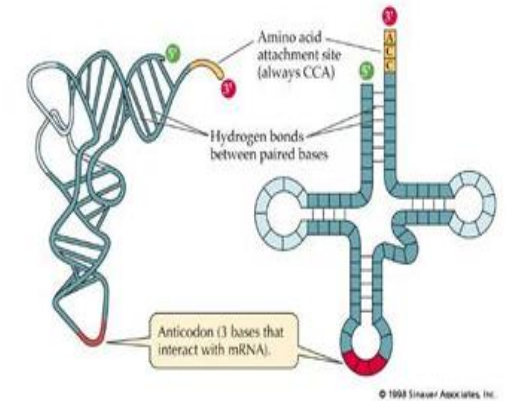
RNA

- Ribonucleic acid or RNA is a vital molecule with a long chain of nucleotides.
- A nucleotide chain comprises **a phosphate, a ribose sugar, and nitrogenous base**.
- RNA acts as a catalyst and as genetic material. There are two types of RNA, that is genetic and non-genetic.



Messenger RNA

Transfer RNA



Ribosomal RNA

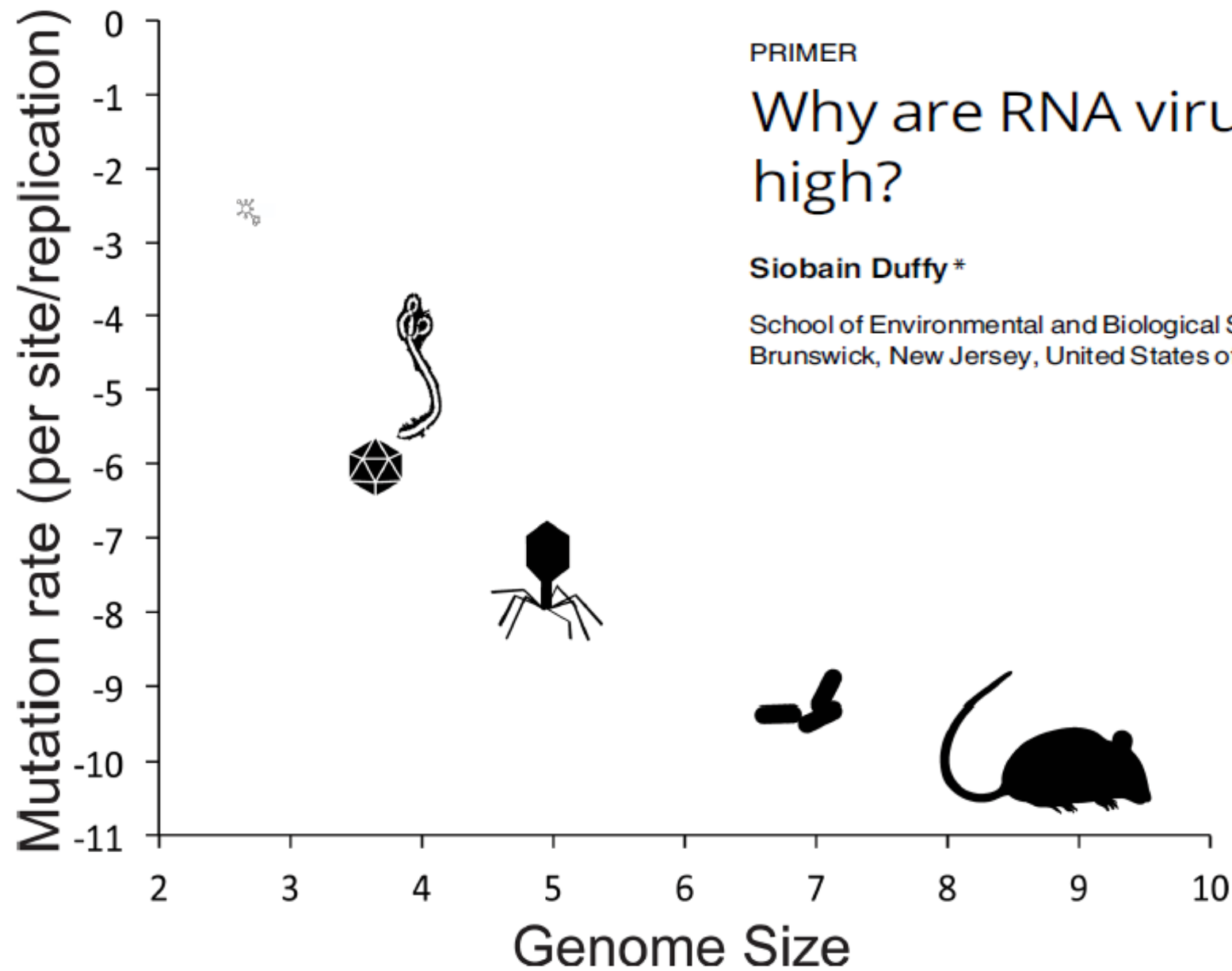
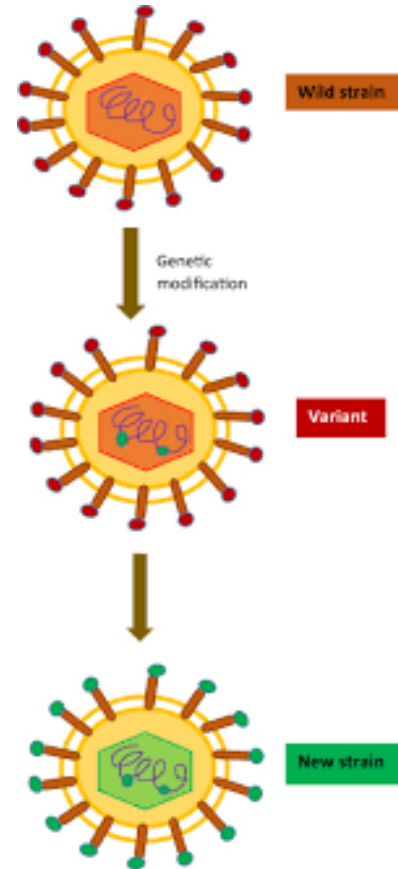


Fig 1. Biological mutation rates summarized from fastest to slowest: Viroid (RNA elements that cause some plant disease without encoding any genes), viruses (RNA shown as Ebola, single-stranded DNA shown as an icosahedron, and double-stranded DNA shown as a myophage), prokaryotes (rod-shaped bacteria), and eukaryotes (rodent). Icons are roughly the size of the range of mutation rates and genome sizes of measured organisms within that group. Axes are log-transformed, data as in [3]. Images are in the public domain except viroid [4], single-stranded DNA virus (icon made by Pixel perfect, www.flaticon.com), and rodent (icon made by Freevik, www.flaticon.com).



Why Do Viruses Mutate?

Mutations allow viruses to:



Move from host
to host more
effectively



Reproduce
faster



Adhere better to
the surfaces of
target host cells



Evade our
immune systems
and vaccines

Sources: Pfizer, UnityPoint Health

DNA or RNA?

A molecule that can act as a genetic material must fulfill the following criteria:

- It should be able to generate its replica (Replication).
- It should be stable chemically and structurally.
- It should provide the scope for slow changes (mutation) that are required for evolution.
- It should be able to express itself in the form of 'Mendelian Characters'.

Mendelian characters are traits (like flower color, seed shape, eye color, etc.) that are passed from parents to offspring according to **predictable patterns**, like:

1. **Dominant** traits — show up if even one parent passes it (e.g., tall plant).
2. **Recessive** traits — show up only if both parents pass it (e.g., short plant).

DNA or RNA?

- If one examines each requirement one by one, because of rule of base pairing and complementarity, both the nucleic acids (DNA and RNA) have the ability to direct their duplications. The other molecules in the living system, such as proteins fail to fulfill first criteria itself.
- **DNA chemically is less reactive and structurally more stable** when compared to RNA. Therefore, among the two nucleic acids, the **DNA is a better genetic material**.
- The presence of thymine at the place of uracil also confers additional stability to DNA.
- Both DNA and RNA are able to mutate. In fact, RNA being unstable, mutate at a faster rate.
- RNA can directly code for the synthesis of proteins, hence can easily express the characters. DNA, however, is dependent on RNA for synthesis of proteins.
- DNA being more stable is preferred for storage of genetic information. For the transmission of genetic information, RNA is better.

Genetic code

- The genetic code can be defined as a set of rules wherein the information encoded in genetic materials are translated into proteins by living cells. **The code defines how codons specify which amino acids will be added next during protein synthesis.**
- The genetic code can also be defined as a relationship between the sequence of amino acids in a nucleotide chain of mRNA or DNA and amino acid in a polypeptide chain.

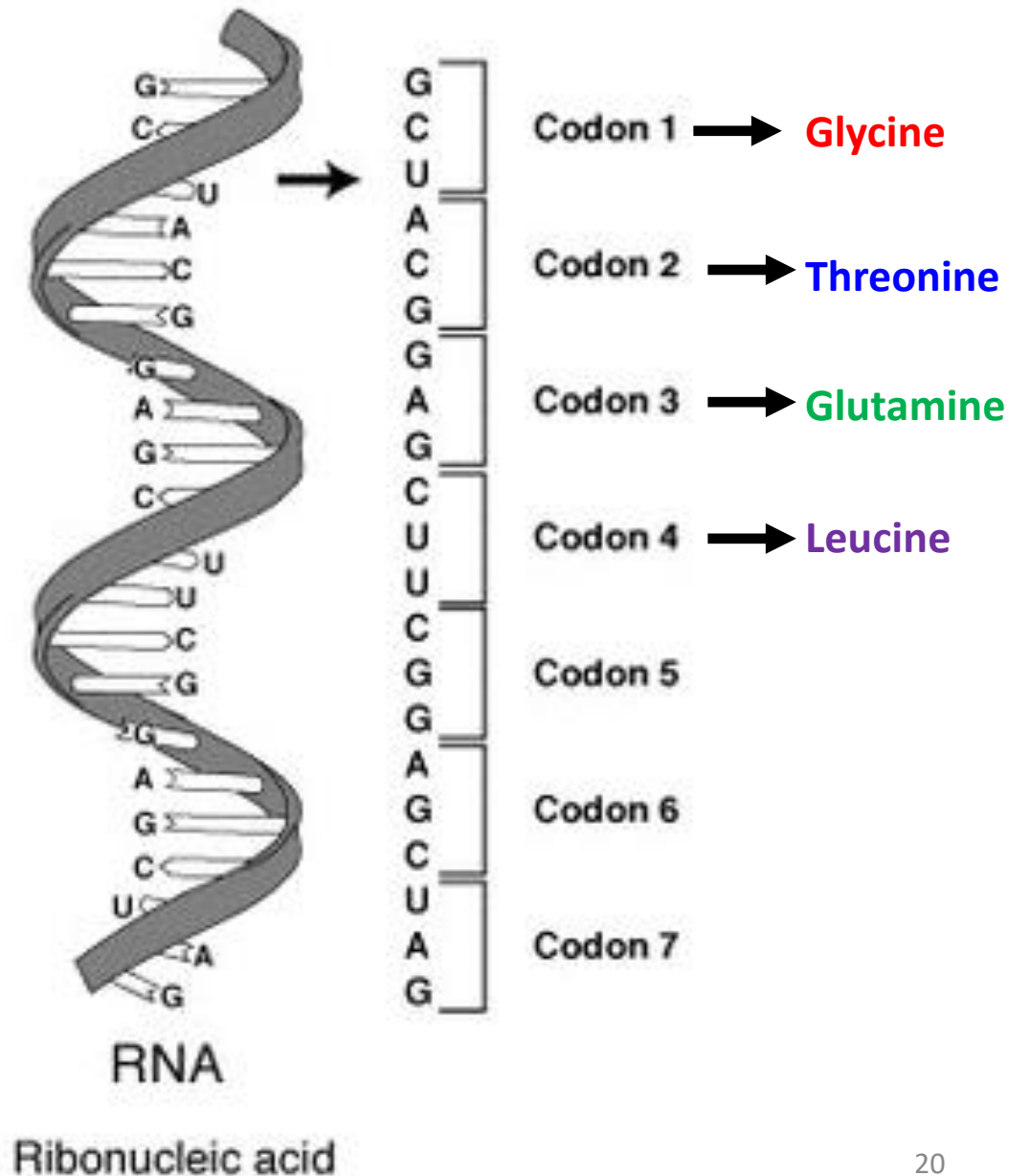
		Second letter				
		U	C	A	G	
First letter	U	UUU Phe UUC UUA Leu UUG	UCU UCC Ser UCA UCG	UAU Tyr UAC UAA STOP UAG STOP	UGU Cys UGC UGA STOP UGG Trp	U C A G
	C	CUU CUC Leu CUA CUG	CCU CCC Pro CCA CCG	CAU His CAC CAA Gln CAG	CGU CGC Arg CGA CGG	U C A G
	A	AUU AUC Ile AUA AUG Met	ACU ACC Thr ACA ACG	AAU Asn AAC AAA Lys AAG	AGU Ser AGC AGA Arg AGG	U C A G
	G	GUU GUC Val GUA GUG	GCU GCC Ala GCA GCG	GAU Asp GAC GAA Glu GAG	GGU GGC Gly GGA GGG	U C A G

Genetic code

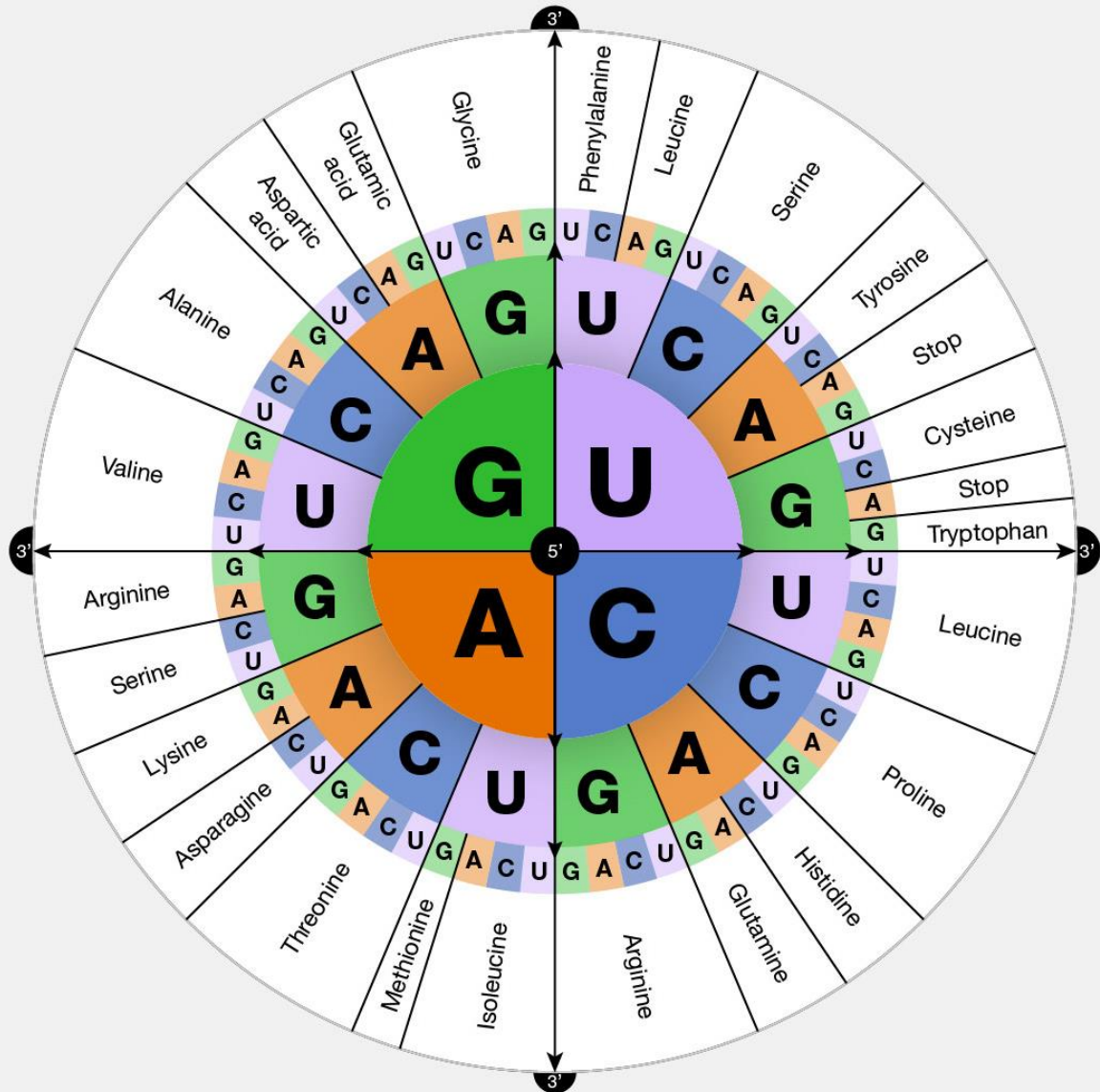
Nearly twenty types of amino acids participate in protein synthesis. **Sixty-one codons out of sixty-four codons code only for amino acids.**

The characteristics of the genetic code are stated below:

- **Degeneracy of genetic code.**
- **Non-overlapping.**
- **Universality.**
- **Triplet in nature.**
- **Comma-less.**
- **Non-ambiguous.**



Genetic code



- Genetic code refers to the instructions contained in a gene that tell a cell how to make a specific protein.
- Each gene's code uses the four nucleotide bases of DNA: adenine (A), cytosine (C), guanine (G) and thymine (T) — in various ways to spell out three-letter “codons” **that specify which amino acid is needed at each position within a protein.**

Chapter-04:

Genomics and Genetics:

Ref book: Biology for Engineers - Arthur T. Johnson [2nd edition]
Biology for Engineers – G. K. Suraishkumar
and

https://www.youtube.com/watch?v=o_-6JXLYS-k

BIO-3105, Biology for Engineers
Lecture-7; 01/04/2026
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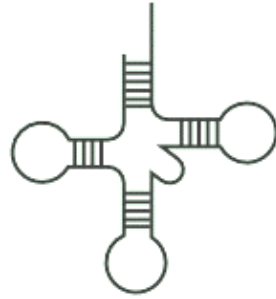
RNA

mRNA



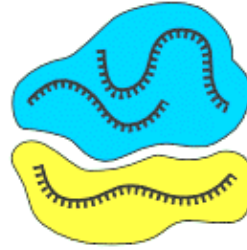
Encodes Proteins

tRNA



Act as adaptor between mRNA and amino acids

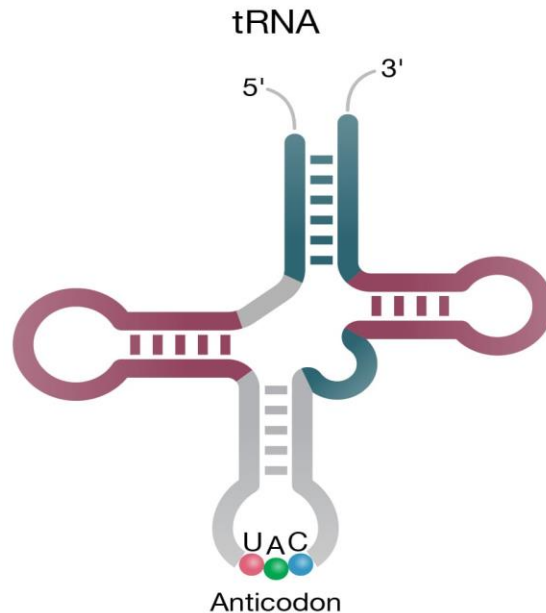
rRNA



Forms the ribosome

Ribonucleic acid or RNA is a vital molecule with a long chain of nucleotides. It is the first genetic material.

A nucleotide chain comprises a phosphate, a ribose sugar, and nitrogenous base. RNA acts as a catalyst and as genetic material for protein synthesis.



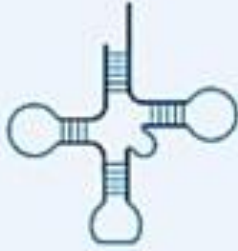
Types of RNA

mRNA



Encodes proteins

tRNA



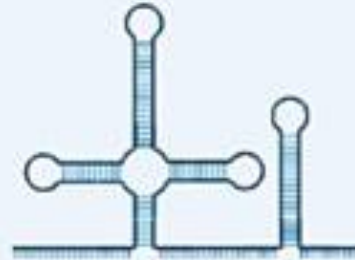
Acts as adaptor between mRNA and amino acids

rRNA



Forms the ribosome

snRNA



Functions in various nuclear processes (e.g. splicing)

snoRNA



Facilitates chemical modification of RNAs

miRNA



Regulates gene expression

siRNA



Silences gene expression

lncRNA

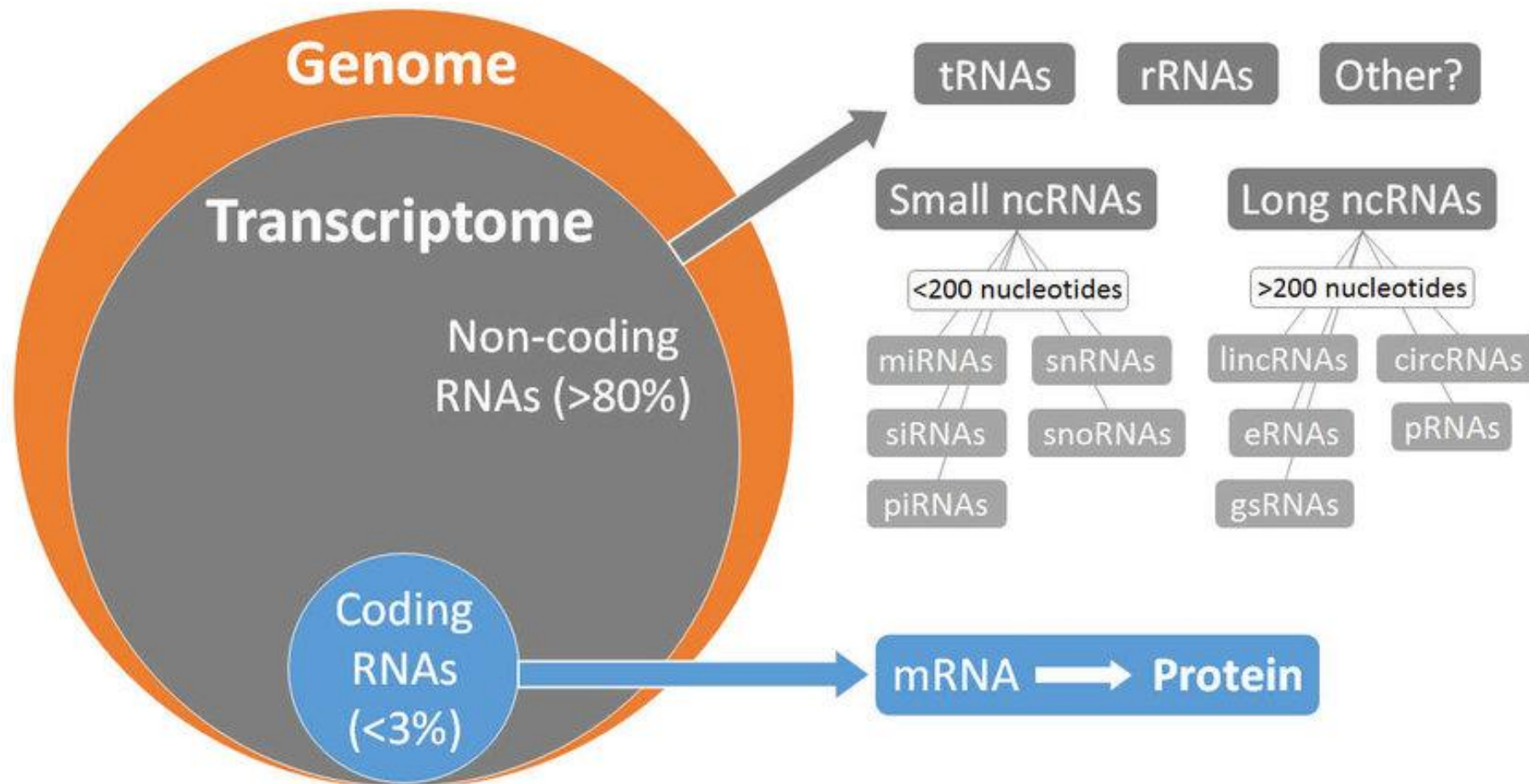


Regulates gene expression

Regulation of Gene Expression

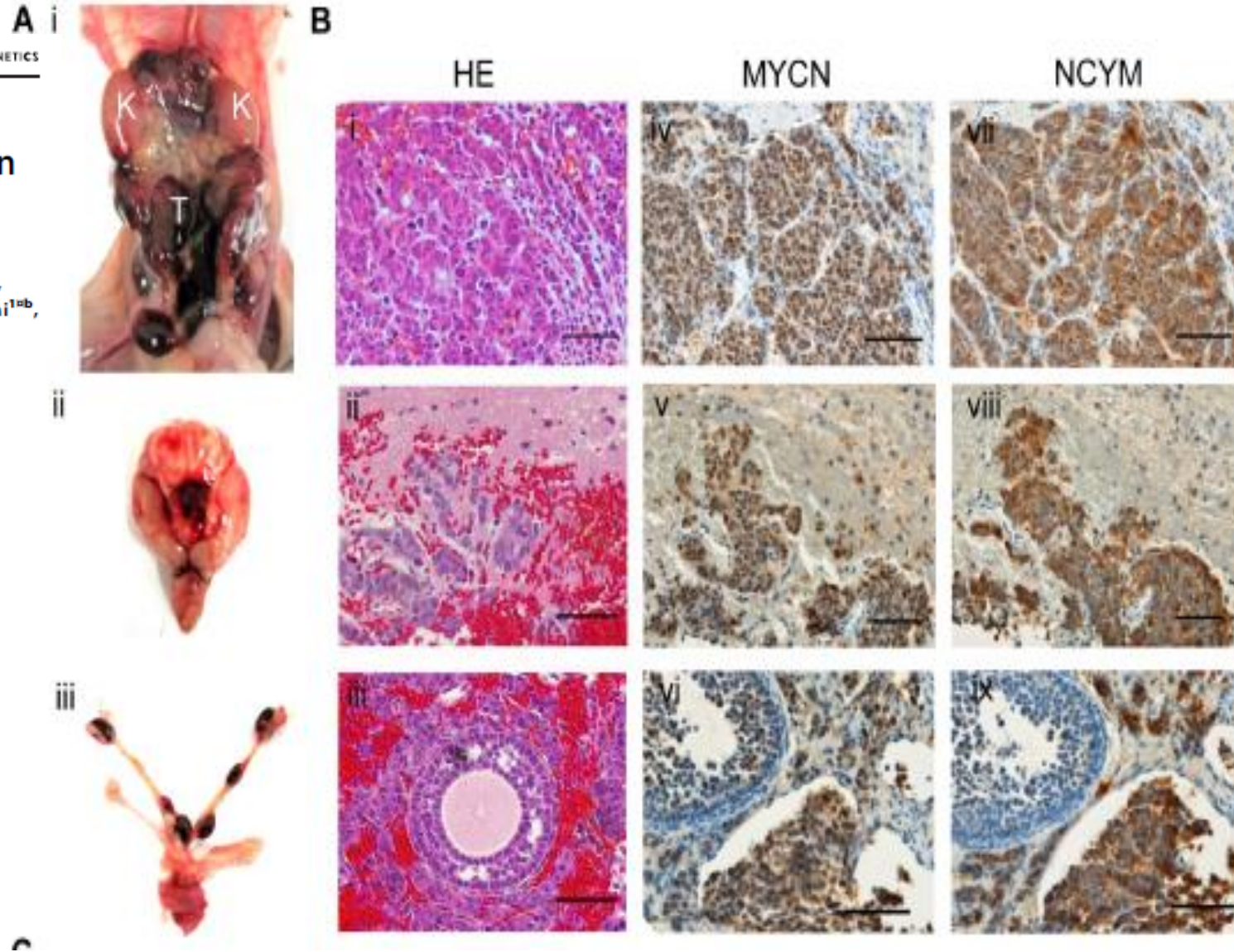
Coding and Non coding RNAs

- **Coding RNAs** generally refers to *mRNA that encodes protein* to act as various components including enzymes, cell structures, and signal transducers.
- **Non-coding RNAs** act as *cellular regulators without encoding proteins*.



NCYM, a *Cis*-Antisense Gene of *MYCN*, Encodes a *De Novo* Evolved Protein That Inhibits GSK3 β Resulting in the Stabilization of MYCN in Human Neuroblastomas

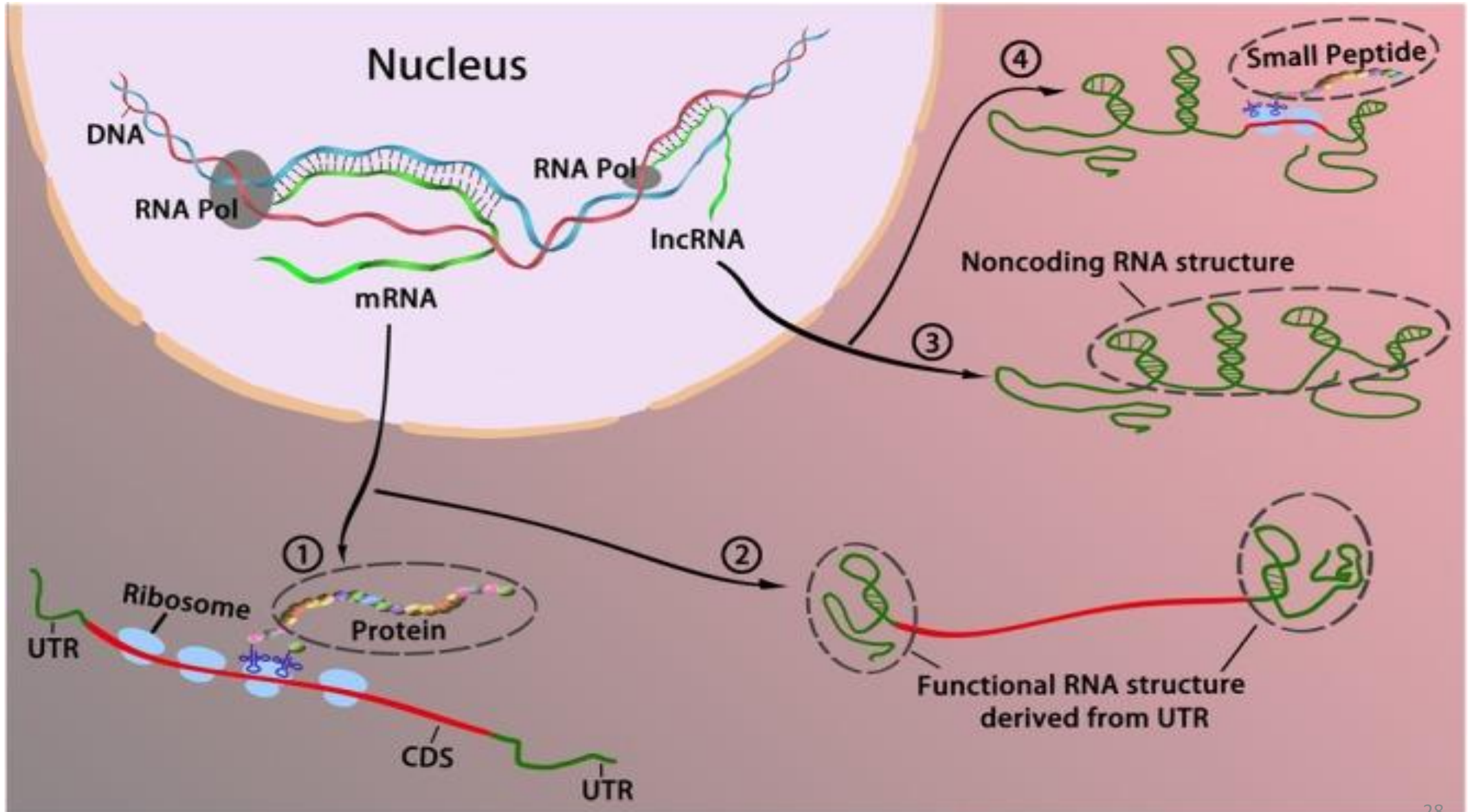
Yusuke Suenaga¹, S. M. Rafiqul Islam¹, Jennifer Alagu¹, Yoshiki Kaneko¹, Mamoru Kato², Yukichi Tanaka³, Hidetada Kawana⁴, Shamim Hossain^{1,2a}, Daisuke Matsumoto¹, Mami Yamamoto¹, Wataru Shoji^{1,5}, Makiko Itami⁴, Tatsuhiro Shibata², Yohko Nakamura¹, Miki Ohira⁶, Seiki Haraguchi^{1,2b}, Atsushi Takatori¹, Akira Nakagawara^{1*}



Coding and Non coding RNAs

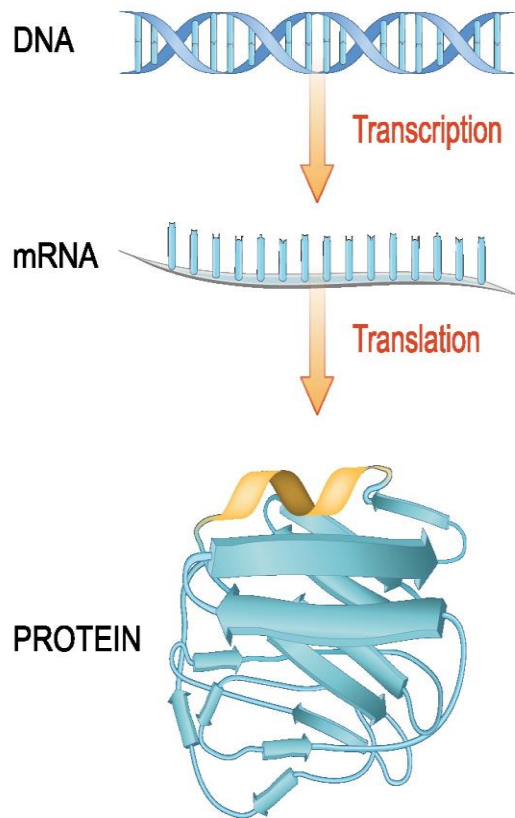
Coding DNA	Noncoding RNA
They have a DNA sequence to encode for proteins.	These DNA sequences do not encode for protein.
The coding sequence has exons.	It has regulatory elements, introns, repeating sequences, pseudogenes, and telomeres.
They are approximately 1% of our total genome.	They form around 99% of our genome.
They encode proteins that have structural , as well as functional importance.	They control and regulate gene activity.

Coding and Non coding RNAs



Different types of RNAs:

Transcription and Translation



There are several different types of RNAs with different structures and functions. These include but are not limited to mRNA, miRNA, rRNA, tRNA, snRNA, snoRNA, lncRNA, and cRNA (Ribozymes). They are typically separated into two categories, **protein-coding and non-coding**. Currently, mRNA and miRNA molecules are most commonly used for RNA.

messenger RNA (mRNA) – is the RNA that carries information from DNA to the ribosomes (site of protein synthesis) in the cell. The mRNA code sequences determine the amino acid sequence in the protein that is produced. These are considered **protein-coding RNA** molecules and can vary in length.

microRNA (miRNA) – as the name implies, are small RNA molecules consisting of ~22 nucleotides. We continue to learn about the exact roll of miRNAs as they are used by the body to regulate **expression of mRNA molecules**. These are considered **non-coding RNA** molecules.

THE NOBEL PRIZE IN PHYSIOLOGY OR MEDICINE 2024

Illustrations: Niklas Elmehed



Victor Ambros

Gary Ruvkun

"for the discovery of microRNA and its role
in post-transcriptional gene regulation"

THE NOBEL ASSEMBLY AT KAROLINSKA INSTITUTET

nobelprize_org BREAKING NEWS

The Nobel Assembly at the Karolinska Institutet has tod...

Cell

Volume 75, Issue 5, 3 December 1993, Pages 843-8

Article

The *C. elegans* heterochronic gene 4 encodes small RNA with antisense complementarity to *lin-14*

Rosalind C. Lee ^{★†}, Rhonda L. Feinbaum ^{★†}, Victor Ambros [†]

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[https://doi.org/10.1016/0092-8674\(93\)90529-Y](https://doi.org/10.1016/0092-8674(93)90529-Y)

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Abstract

lin-4 is essential for the normal temporal control of diverse postembryonic developmental events in *C. elegans*. *lin-4* by negatively regulating the level of LIN protein, creating a temporal decrease in LIN protein starting in the first larval stage. We have cloned the *C. elegans lin-4* locus by chromosomal walking and transformational rescue. We used the *C. elegans* clone to identify the gene from three other *Caenorhabditis* species; all four *Caenorhabditis* clones

Cell

Volume 75, Issue 5, 3 December 1993, Pages 855-862

Article

Posttranscriptional regulation of the heterochronic gene *lin-14* by *lin-4* mediates temporal pattern formation in *C. elegans*

Bruce Wightman ^{★†}, Ilho Ha [★], Gary Ruvkun

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[https://doi.org/10.1016/0092-8674\(93\)90530-4](https://doi.org/10.1016/0092-8674(93)90530-4)

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Abstract

During *C. elegans* development, the temporal pattern of many cell lineages is specified by graded activity of the heterochronic gene *Lin-14*. Here we demonstrate that a temporal gradient in Lin-14 protein is generated posttranscriptionally by multiple elements in the *lin-14* 3'UTR that are regulated by the heterochronic gene *Lin-4*. The *lin-14* 3'UTR is both necessary and sufficient to confer *lin-4*-mediated posttranscriptional temporal regulation. The function of the *lin-14* 3'UTR is conserved between *C. elegans* and *C. briggsae*.

microRNA (miRNA, μ RNA)

MicroRNA (miRNA) is a small, non-coding RNA molecule that regulates gene expression in cells:

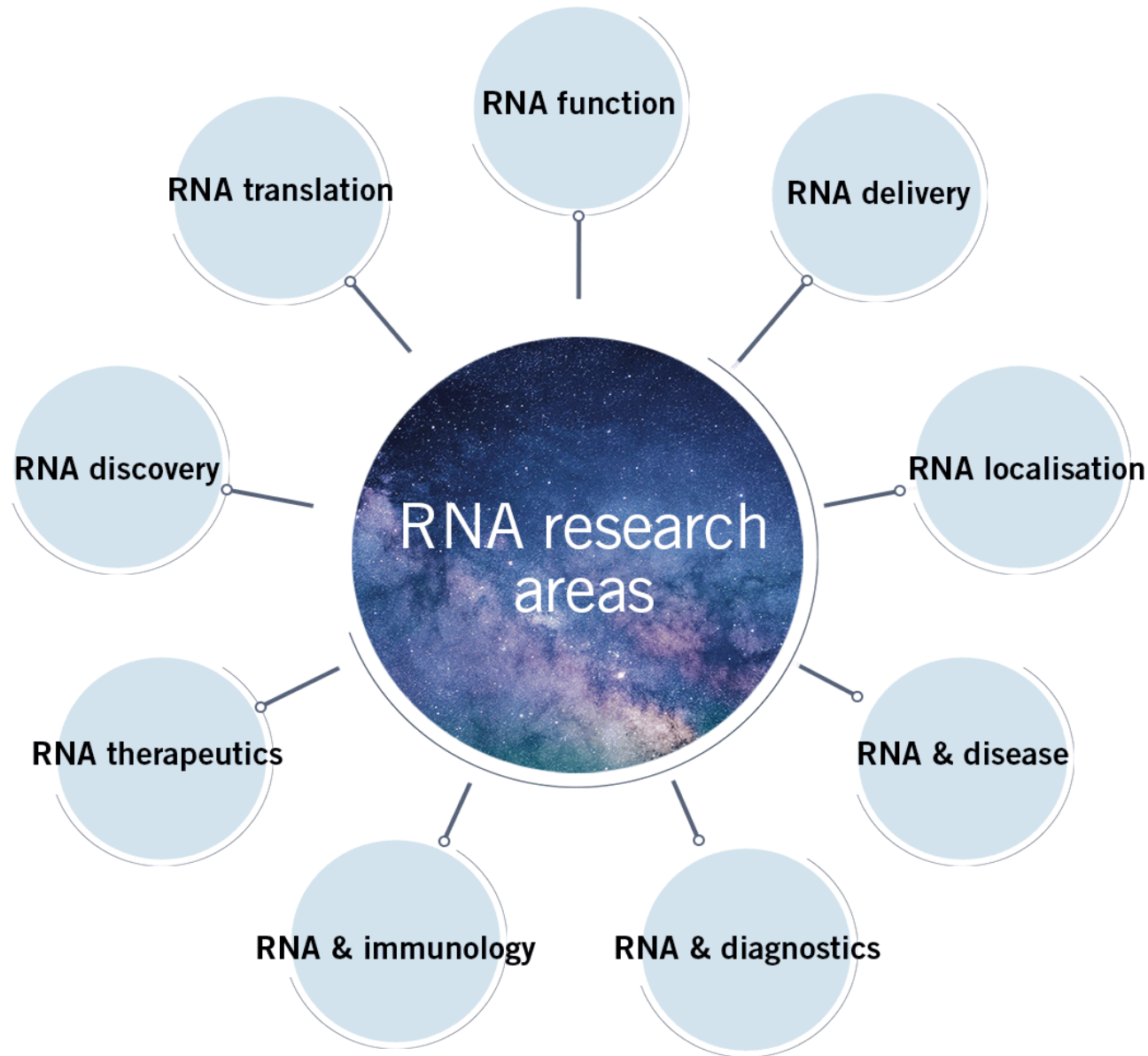
- **Function:** miRNAs control the types and amounts of proteins cells make by binding to messenger RNA (mRNA) in the cytoplasm. This binding prevents or delays translation.
- **Length:** miRNAs are about 22 nucleotides long.
- **Location:** miRNAs are encoded in the genomes of plants and animals.
- **Discovery:** Victor Ambros' laboratory discovered miRNAs in 1993 in *Caenorhabditis elegans*.
- **Significance:** The discovery of miRNAs is considered a major landmark in modern molecular biology.
- **Role in disease:** miRNAs are associated with many diseases, including cancer, cataract, progressive hearing loss, and skeletal and growth defects.
- **Role in drug resistance:** miRNAs are linked to drug resistance mechanisms in tumors.
- **Role in viral disease:** Viral miRNAs play an important role in the pathogenesis of viral disease.

RNA therapeutics

- RNA therapeutics are a new class of medications based on ribonucleic acid (RNA). Research has been working on clinical use since the 1990s, with significant success in cancer therapy in the early 2010s.
- In 2020 and 2021, mRNA vaccines have been developed globally for use in combating the coronavirus disease (COVID-19 pandemic).
- The Pfizer–BioNTech COVID-19 vaccine was the first mRNA vaccine approved by a medicines regulator, followed by the Moderna COVID-19 vaccine, and others.

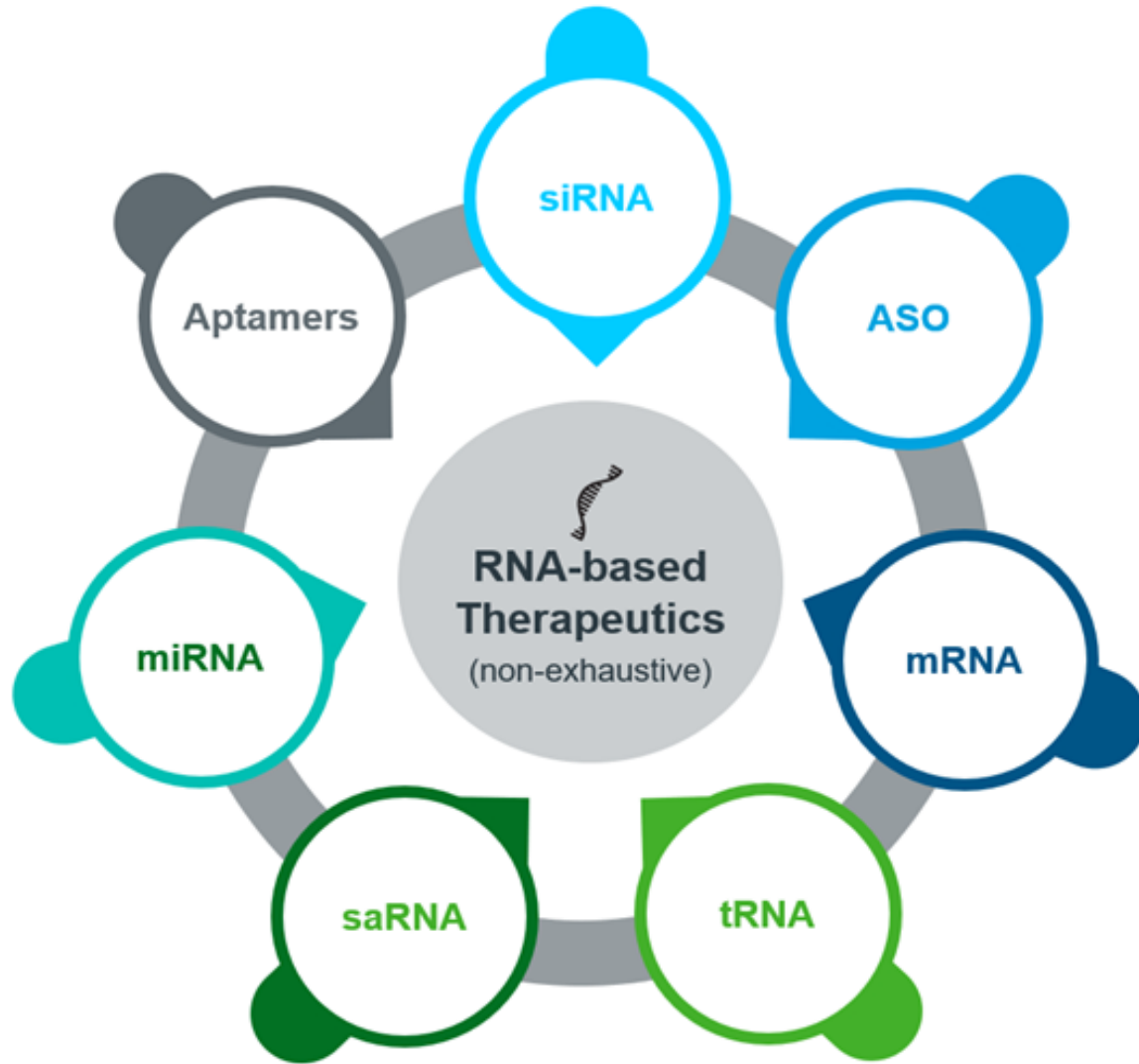
ABSTRACT: “RNA therapeutics” refers to a disease treatment or drug that utilizes RNA as a component. In this context, RNA may be the direct target of a small-molecule drug or RNA itself may be the drug, designed to bind to a protein, or to mimic or target another RNA. RNA has gained attention in the drug-development world, as recent clinical successes and breakthrough technologies have revolutionized the drug-like qualities of the molecule or its usefulness as a drug target. In this special issue of RNA, we gathered expert perspectives on the past, present, and future of the field, to serve as a primer and also a challenge to the broad scientific community to incorporate RNA into their experimental design and problem-solving process, and to imagine and realize the potential of RNA as a therapeutic drug or target.

Future of RNA in research



Find your Future

RNA has many flavors

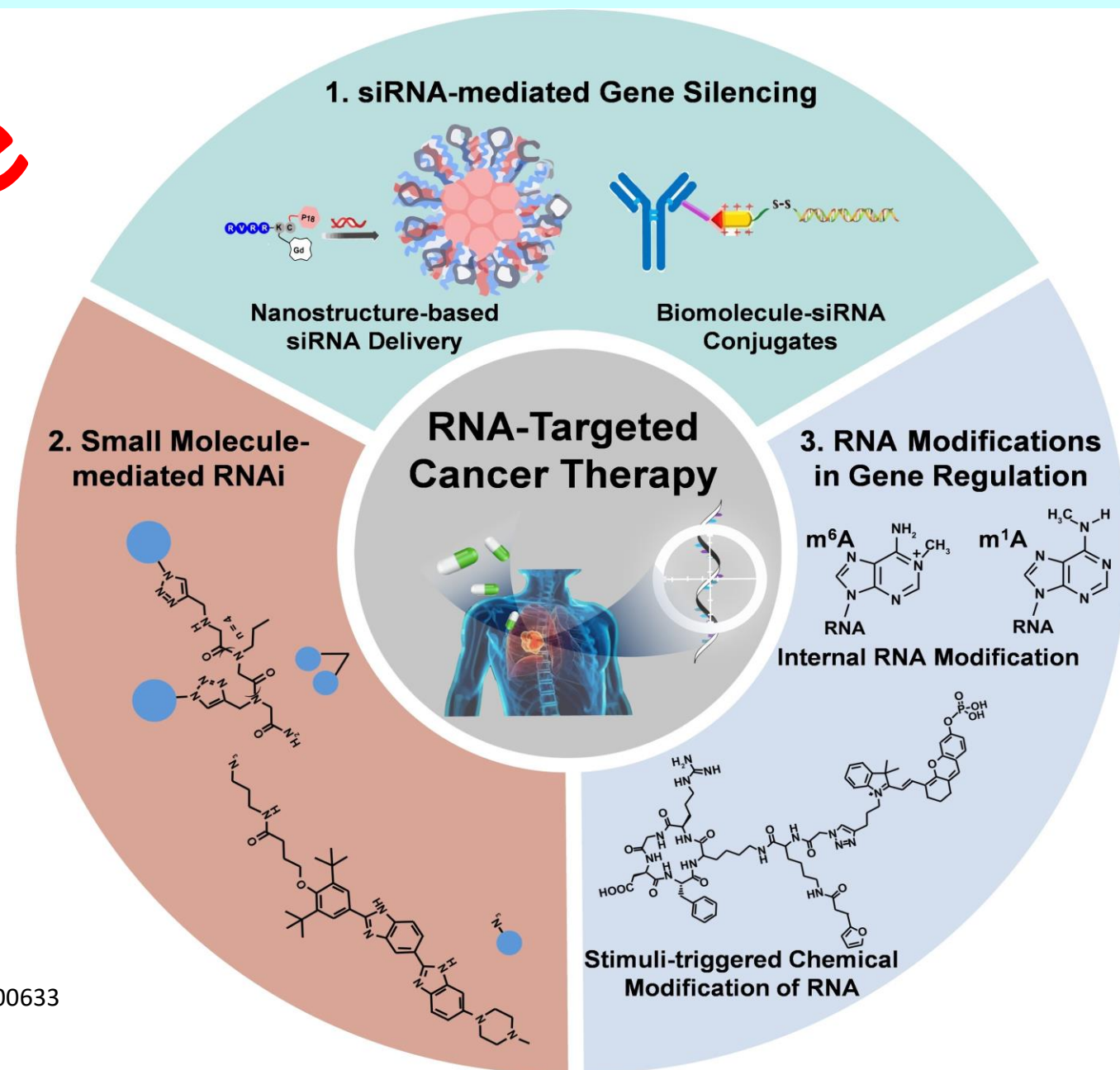


RNA type	Mechanism of action	Details
siRNA (small interfering RNA)	Pathway interference	RNA designed to target a specific mRNA for degradation, resulting in gene silencing.
ASO (antisense oligonucleotide)	3D-function	RNA which prevents mRNA physically interacting with proteins involved in splicing or translation.
mRNA (messenger RNA)	Replacement	mRNA can be a replacement for one that is not expressed correctly, or mimic an antigen expressed by a pathogen.
tRNA (transfer RNA)	Replacement	tRNA therapies seek to overcome mutations that result in truncated proteins.
saRNA (self-amplifying RNA)	Replacement	saRNA allows for much greater expression of the protein it seeks to produce.
miRNA (microRNA)	Pathway interference	miRNA therapies inject miRNA that is under-expressed due to genetic mutations.
Aptamers	Pathway interference	Aptamers are molecules that bind to a specific protein, directly moderating protein-protein interactions.

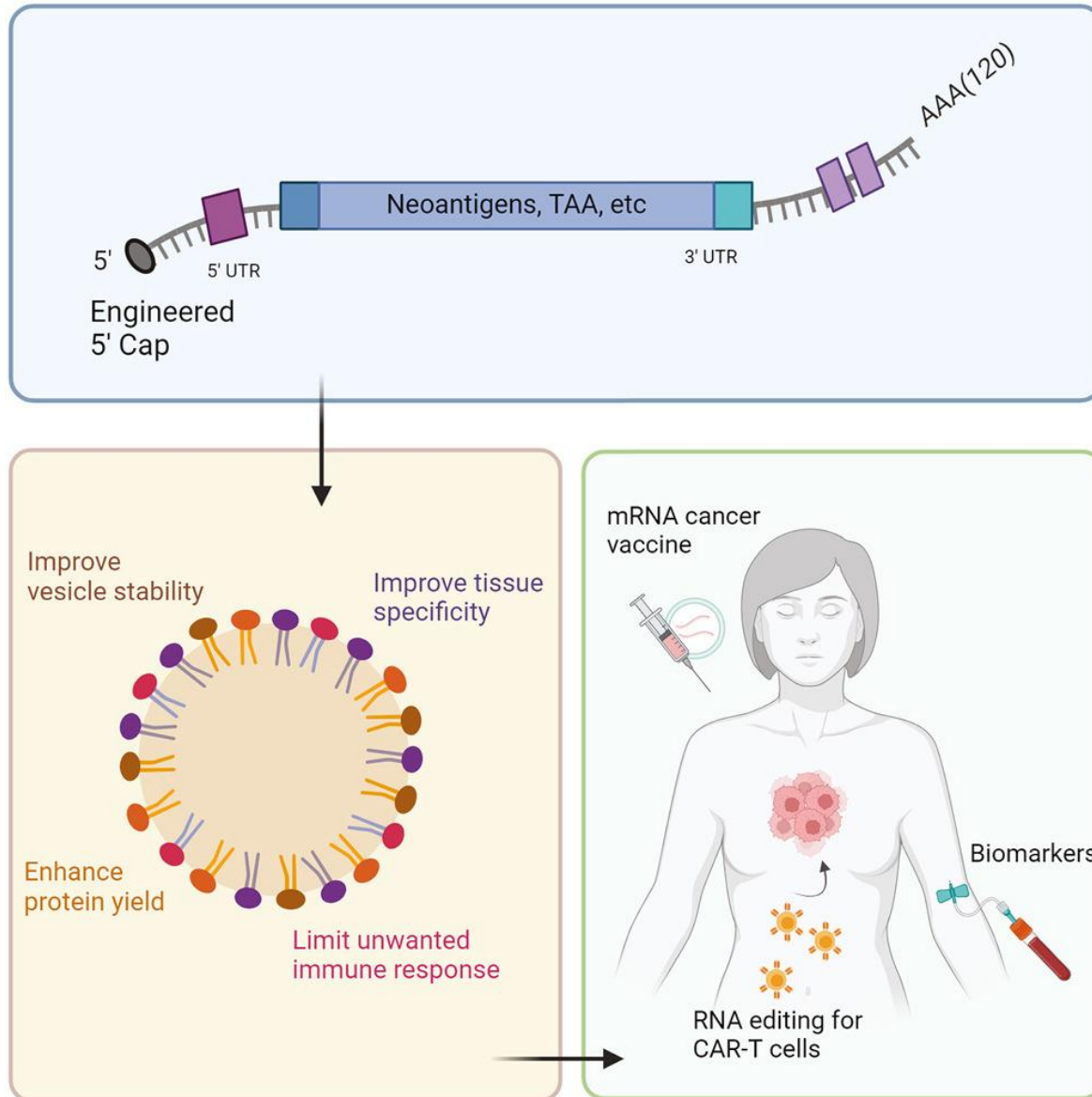
Find your Future

RNA therapeutics in cancer treatment

Find your Future



RNA therapeutics in Cancer Treatment



Find your Future

RNA induced immunity/memory

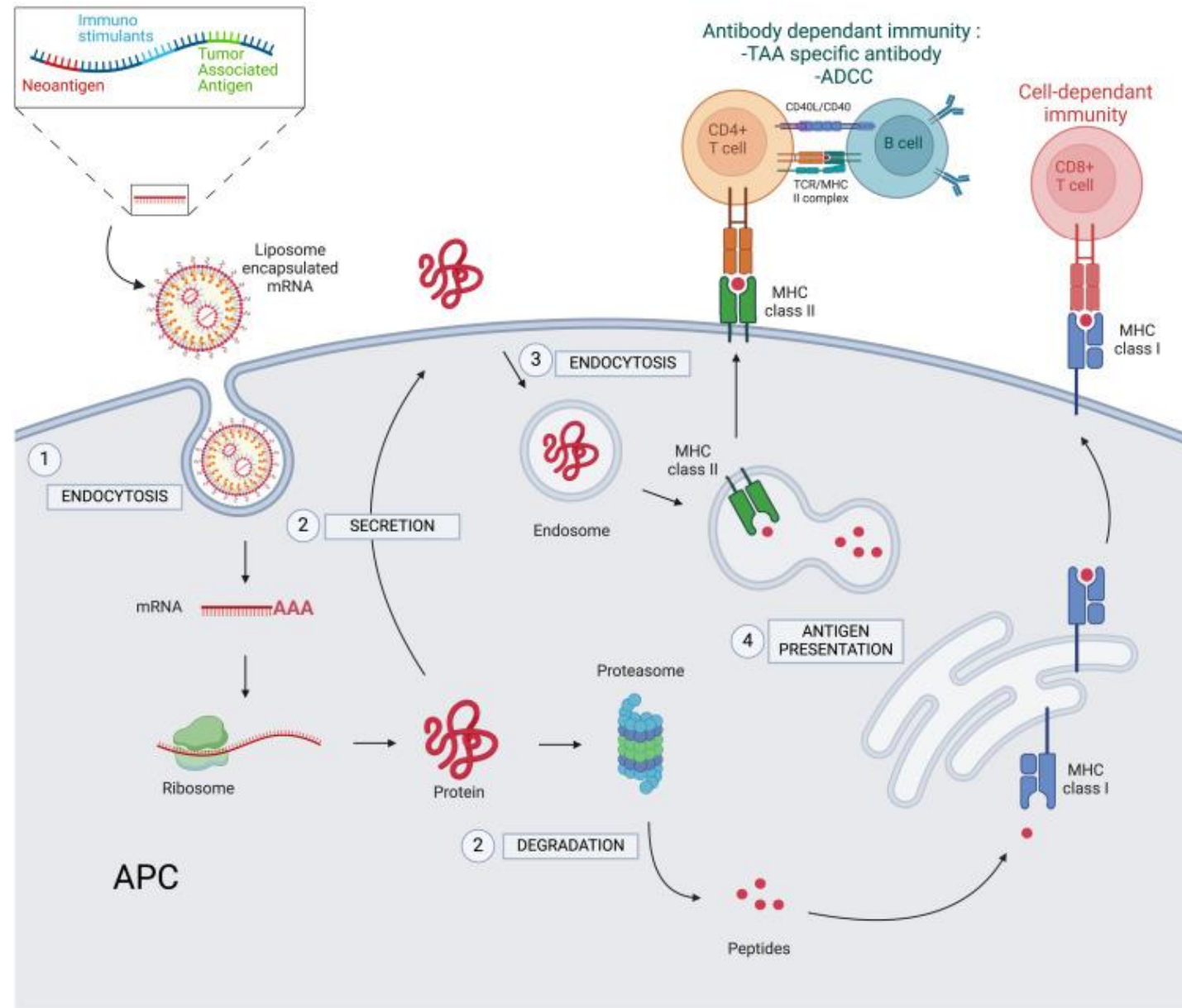
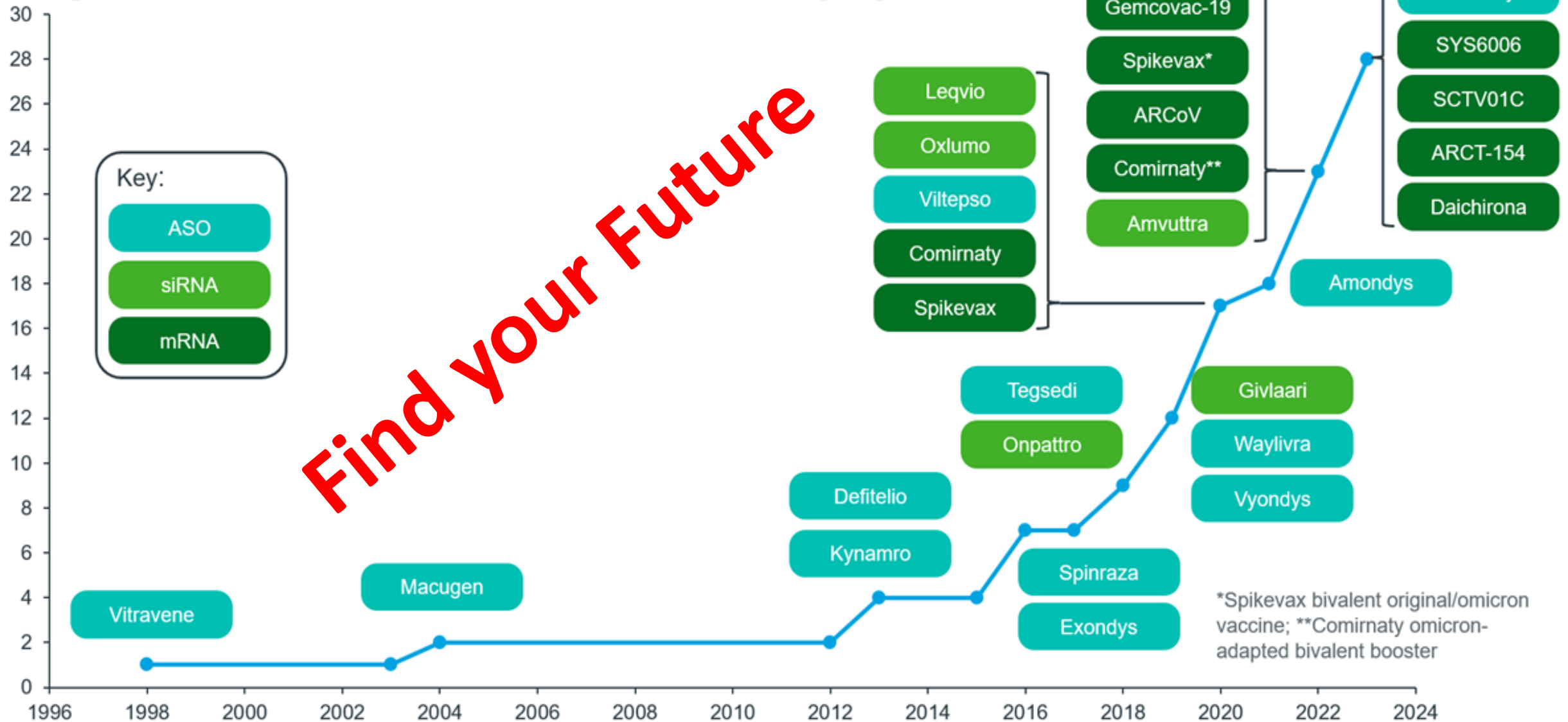
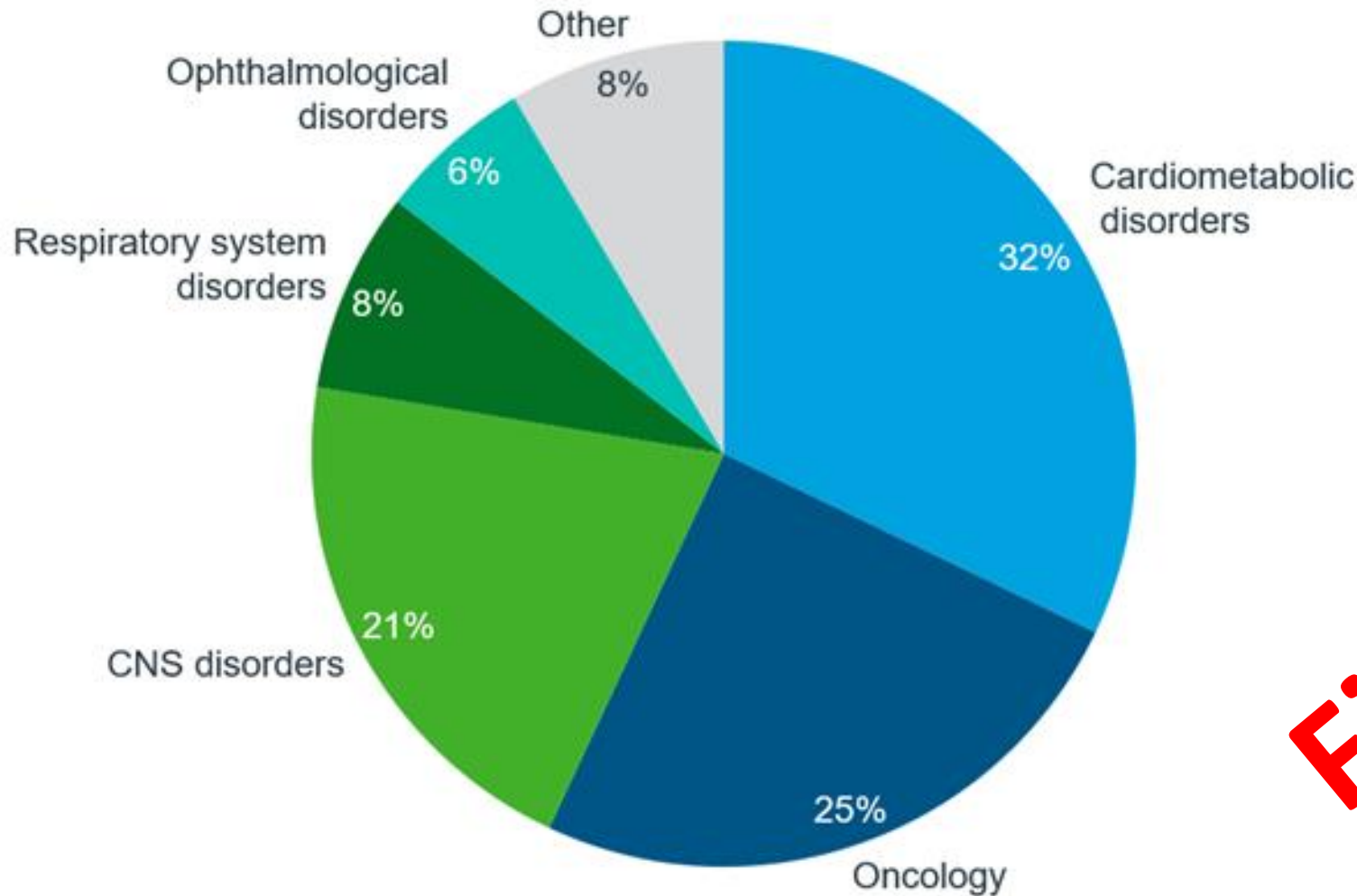


Figure 1: RNA products approved by type



Therapeutic arena of RNA therapeutics



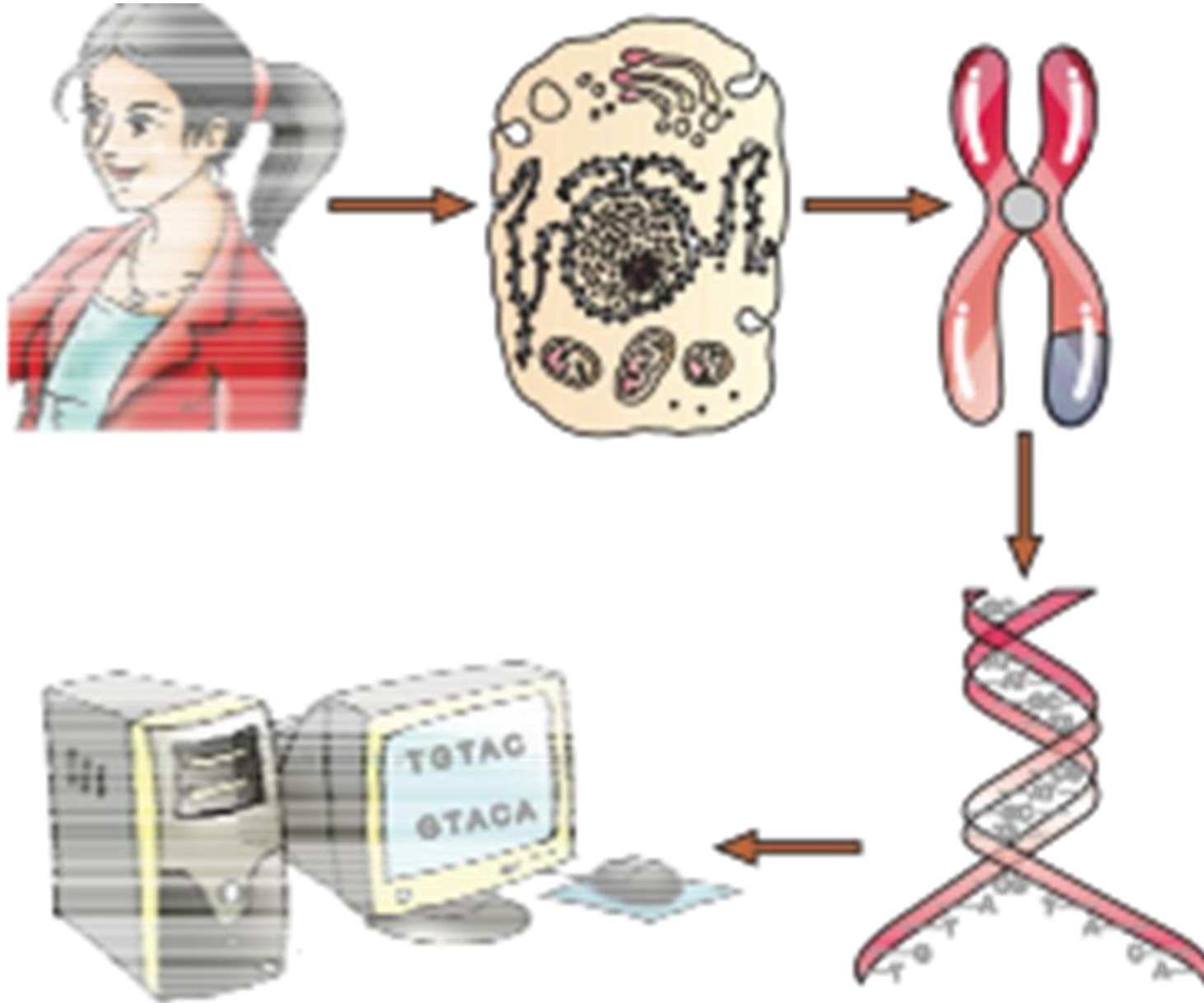
Find your Future



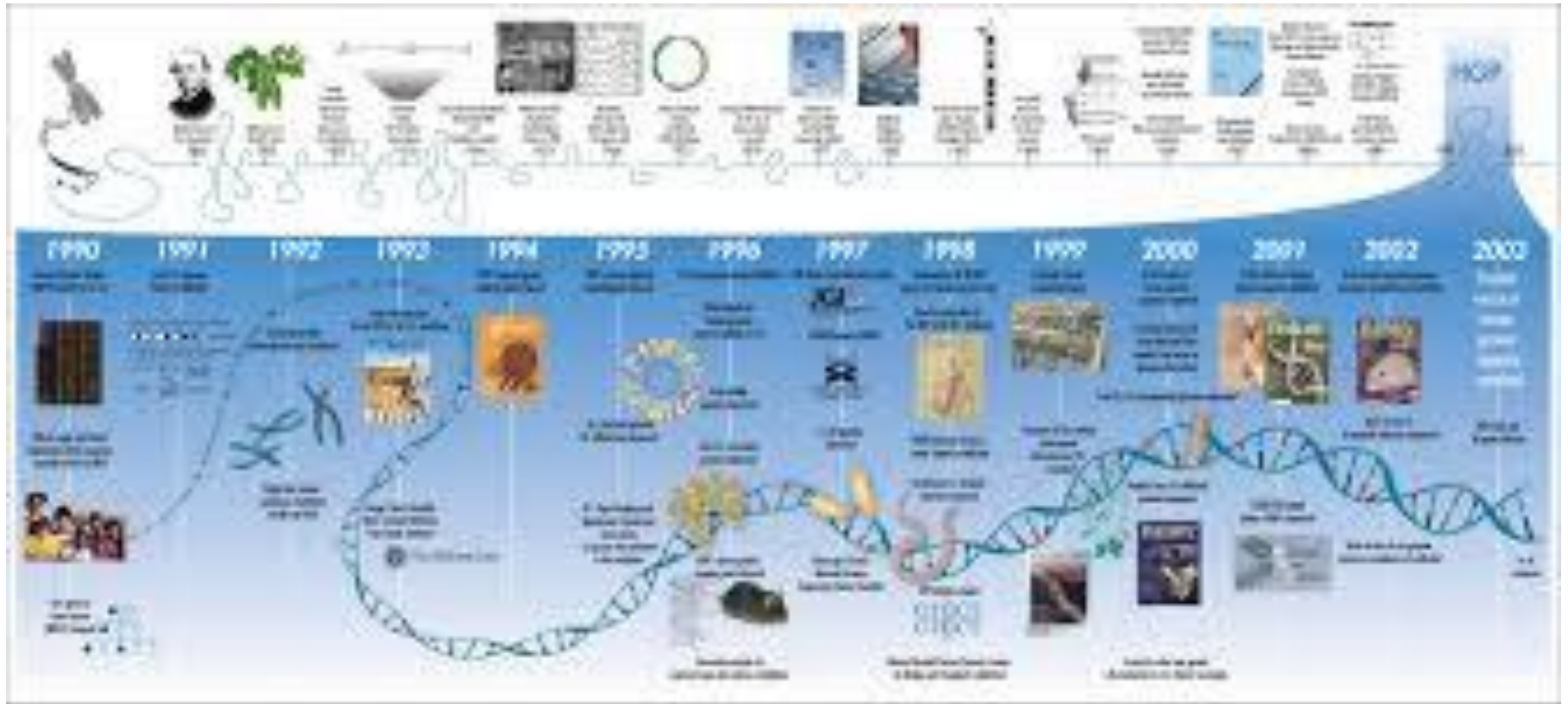
National Human Genome Research Institute

<https://www.genome.gov/genetics-glossary/Genetic-Code>

Human Genome Project



History and timeline of Human Genome Project (HGP)



https://www.genome.gov/sites/default/files/media/files/2020-09/HGP_Timeline.pdf

Human Genome Project

The human genome project was launched to sequence the entire human genome of 2.75 billion base pairs. The main goals of the human genome project are:

- To provide a complete sequence of 3 billion base pairs that make up the human genome.
- To sequence the genome of other organisms that are used in medical research; e.g., mouse, flies, monkeys, etc.
- To develop new tools to obtain and analyze the data and to make this data widely available.
- It holds prospects for healthier living, a database of knowledge about designer drugs, genetically modified diets, and genetic identity.

The Goal of Human Genome Project:

Some of the important goals of HGP were as follows:

- (i) Identify all the approximately **20,000-25,000 genes** in human DNA;
- (ii) Determine the **sequences of the 3 billion chemical base-pairs** that make up human DNA;
- (iii) Store this information in **databases**;
- (iv) Improve **tools for data analysis**;
- (v) Transfer related technologies to other sectors, **such as industries**;
- (vi) Address the **ethical, legal, and social issues (ELSI)** that may arise from the project.

Find your Future

Human Genome Project (HGP)

Begun in 1990, the HGP was an ambitious endeavor to generate the first sequence of the human genome. We brought *All of Us* staffers together to reflect on how the HGP laid the foundation for *All of Us*.

“With the Human Genome Project, what we were so excited about was having a map of the genes,” said Nicole McNeil Ford, Ph.D., an *All of Us* Program Officer and a former National Human Genome Research Institute (NHGRI) research biologist. “People were familiar with chromosomes, but we didn't really know exactly where most genes were mapped on those chromosomes.”

Genomic medicine

- Genomic medicine is a medical discipline that involves using a [person's genomic information as part of their clinical care](#). Other similar terms include [individualized medicine](#), [personalized medicine](#) and [precision medicine](#).
- Genetic medicines are genetic materials such as [DNA and RNA delivered into the body as a therapeutic](#). They are a promising new class of medicine that was not possible even a short time ago.

Researchers are testing several approaches, including:

- **Replacing a disease-causing mutated gene with a healthy gene**
- **Inactivating, or “knocking out,” a malfunctioning mutated gene**
- **Introducing a new gene to help fight a disease**

<https://www.precisionnanosystems.com/resources-and-community/training-education/what-is-genetic-medicine>

Genomic medicine is the use of an individual's genomic information (DNA sequence, mutations, epigenetic changes) in clinical care.

Involves prevention, diagnosis, prognosis, and treatment tailored to a patient's unique genetic makeup.

Genomic Sequencing: Whole genome, exome, or targeted sequencing.

Bioinformatics: Data analysis, interpretation of genetic variations.

Molecular Diagnostics: Biomarkers, gene panels, companion diagnostics.

Gene Editing Tools: CRISPR-Cas9, base editing, prime editing.

Pharmacogenomics: Understanding drug response based on genetic variations.

Applications:

Risk Prediction: Identifying susceptibility to cancer, cardiovascular, neurological, and metabolic diseases.

Early Diagnosis: Detection of rare genetic disorders, prenatal and neonatal screening.

Targeted Therapies: Use of precision drugs in oncology (e.g., EGFR inhibitors, HER2 blockers).

Gene Therapy & Cell Therapy: Correcting defective genes (e.g., CAR-T therapy, sickle cell treatment).

Infectious Diseases: Tracking pathogen evolution (e.g., COVID-19 variants).

Genome size due to transposable elements

- There are many enormous differences in size in genomes, specially mentioned before in the multicellular eukaryotic genomes.
- The main reason why there is such a big variety of sizes is due to the presence of transposable elements. TEs are known to contribute to a significant change in a cell's mass of DNA.

Genomic alterations

- All the cells of an organism originate from a single cell, so they are expected to have identical genomes; however, in some cases, differences arise.
- Both the process of copying DNA during cell division and **exposure to environmental mutagens** can result in mutations in somatic cells. In some cases, such mutations lead to cancer because they cause cells to divide more quickly and invade surrounding tissues.
- In certain lymphocytes in the human immune system, **V(D)J recombination generates** different genomic sequences such that each cell **produces a unique antibody or T cell receptors**.
- During meiosis, diploid cells divide twice to produce haploid germ cells. During this process, recombination results in a reshuffling of the genetic material from homologous chromosomes so each gamete has a unique genome.

Genome evolution

- Genomes are more than the sum of an organism's genes and have traits that may be measured and studied without reference to the details of any particular genes and their products. Researchers compare traits such as karyotype (chromosome number), **genome size, gene order, codon usage bias, and GC-content** to determine what mechanisms could have produced the great variety of genomes that exist today.
- Duplications play a major role in shaping the genome. Duplication may range from extension of **short tandem repeats, to duplication of a cluster of genes**, and all the way to duplication of entire chromosomes or even entire genomes. Such duplications are probably fundamental to the **creation of genetic novelty**.

Genome evolution

- Horizontal gene transfer is invoked to explain how there is often an extreme similarity between small portions of the genomes of two organisms that are otherwise very distantly related. **Horizontal gene transfer seems to be common among many microbes.** Also, eukaryotic cells seem to have experienced a transfer of some genetic material from their **chloroplast and mitochondrial genomes to their nuclear chromosomes.** Recent empirical data suggest an important role of viruses and sub-viral RNA-networks to represent a main driving role to generate genetic novelty and natural genome editing.

DNA

Taken the distance between two consecutive base pairs as 0.34 nm (0.34×10^{-9} m), if the length of DNA double helix in a typical mammalian cell is calculated (simply by multiplying the total number of bp with distance between two consecutive bp, that is, 6.6×10^9 bp $\times$ 0.34×10^{-9} m/bp), it comes out to be approximately 2.2 metres. A length that is far greater than the dimension of a typical nucleus (approximately 10^{-6} m). How is such a long polymer packaged in a cell?

DNA Packaging

The negatively charged DNA is packaged by surrounding the positively charged histone octamer. A structure called **nucleosome** is formed. The DNA is packed in chromatin of eukaryotes.